The Norns III: Seeding Simulated-Annealing Placement with Provenance and Recovered Clusters How hot can the anneal start before it destroys a good initial placement?

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Abstract—Simulated annealing is the workhorse of FPGA placement, and it starts from an initial placement that the anneal then destroys and rebuilds. What if the initial placement carried real information – the HDL hierarchy a provenance-tracked front end kept, or the clusters a graph analysis recovered from the flat netlist? This paper asks whether such seeds improve placement, and, because the anneal's early phase can erase any seed, at what initial temperature the seed's advantage dies. The seeds come from Odin III's provenance hierarchy and from Loki III's L2 clusterings (Louvain, spectral, affinity propagation, k-means); the die-level arrangement of clusters comes from a cut-weighted quotient-graph layout; and the seeds enter VPR through its initial-placement and floorplanning-constraint interfaces, with VPR's stock checkpointing flag (on by default, following RLPlace) enabled throughout and an off sweep as the control arm. We report seed retention at scale 1 and at twice each design's crossover, quality of results against an unseeded baseline, and the largest tested temperature scale at which each seed still helps – the recipe a clustering-based initial placement should be used with. A second round adds bitwise seeds (Odin III word membership laid out as bit-sliced runs), a random start and a coarse-random arm, and two structure analyses that need no placement run; each hypothesis is reported with its deciding number, and an appendix keeps the research log. Rounds one and two are over 15 designs with one VPR seed per configuration. A third round asks whether VPR's packing step dilutes the HDL structure, builds a packer (a copy of VPR's AAPack clusterer with a provenance-affinity gain term) and tests it against the stock packer on 15 designs with three VPR seeds per configuration, adds the top-level I/O buses as words, and reads the annealed placement against provenance; its decision rules were committed before any measurement (one later amendment, to the proximity analysis, is recorded). Affinity packing raises word purity without growing the CLB count, but routed wirelength is slightly worse than the stock packer's and that comparison is inconclusive.

Round four. A log-only census of the packer, fixes to the Norns packer (proposal counting, the affinity join) and a post-route critical-path attribution from Odin III answer what gets clustered: over 15 designs the stock packer's CLB pairs are, by median, 4.0% series-critical, 32.0% shared-input or orthogonal and 64.4% series-noncritical or filler (SN+FL, the non-critical remainder; medians SN 3.2%, FL 57.6%). Peter's illustration was 10/75/15: shape not supported, literal numbers not supported (orthogonal is a lower bound: the words cover about half the LUTs, and the registered rule's "t<0.40" arm is sensitive to it). Near-critical edges are split across CLBs less often than edges in general (H17c supported by the literal rule: median rate ratio 0.652, $p = 0.027$, 9 designs with at least 20 hops; with an n<12 clause, a reading outside the registered rule, inconclusive (n < 12)). Hierarchy versus chip size: H13a supported, H13b inconclusive. The norns packer against stock on critical-path delay: not supported; the stage-by-bit seed: see Table XIX. Found in this round: the round-three affinity files lacked every adder (carry-chain) atom, so round three's affinity results describe a packer that did not see carry chains (Section III-I).

A fifth round builds the yardstick the earlier rounds lacked: 72 generated designs whose optimal clustering is known by construction (blocks that each fill one CLB, wired as chains, meshes, trees and random sparse graphs, 4 to 1024 blocks), in the manner of the PEKO/TPEKO placement benchmarks, at two levels: written directly as a netlist, and as Verilog through Odin III. The optimum is exact for the CLB count only; the cut is an upper bound on the minimum cut, and the wirelength and delay ratios are against the placement VPR produces from the forced block clustering, not against a proven placement optimum. Against that clustering the stock VPR packer's gap is CLB count x1.073, wirelength x1.220, critical-path delay x1.093 at the netlist level and CLB count x1.073, wirelength x1.219, critical-path delay x1.082 at the HDL level; the HDL-aware packer's is CLB count x1.031, wirelength x1.099, critical-path delay x1.032. Status under the literal, pre-data rules: the netlist exactness check (H28) is not supported (37 of 72 designs pass the zero-violation rule; the slack rule chosen after the data passes 72 of 72, labelled a variant) and the HDL yardstick check (H18) is not supported (32 of 72 under the original label rule; the amended rule keeps 63, inconclusive). The HDL-aware packer's gain over the stock packer (wirelength x0.902) is a leak, not evidence of HDL knowledge: it sits in dp (wirelength x0.834), where the word file is the block partition (purity 0.9998 against the stock packer's 0.9501), and is zero in ctl (x1.000; 27 of 27 .net files equal the stock files), so this round is no evidence of an HDL-knowledge gain beyond the leak. Protocol 7886f49.

Index Terms—FPGA placement, simulated annealing, provenance, clustering, VTR

I. Introduction

VPR's simulated-annealing placer starts from an initial placement and anneals it: the schedule begins hot enough

to move anything, and the early phase rearranges the circuit wholesale. The question this paper asks is whether the initial placement can be made to mean something, and how much of that meaning survives the anneal. Gharibian et al. showed that system-level structure recovered from a flat netlist correlates with placement locality [1]–[3]. Odin III keeps the provenance link that makes recovery unnecessary for the hierarchy [1], and Loki III recovers clusters from the flat netlist without a language model. RLPlace [4] showed that the anneal itself can be improved – directed moves chosen by a reinforcement-learning agent and checkpointing that prevents oscillation – starting from VTR's analytic placement. This paper connects those lines: the seed carries the structure, the anneal is measured for how much of it the seed can keep, and the largest tested temperature scale at which the advantage survives is reported as data, not tuned away.

The contributions are: (1) a seeding methodology – provenance hierarchy and four recovered clusterings, arranged on the die by a quotient-graph layout, entering VPR through its initial-placement and constraint interfaces, with VPR's stock checkpointing enabled and its off sweep as the control arm; (2) the crossover sweep, which calibrates the highest useful initial temperature per design against the best placement any seed achieves; (3) the retention and quality numbers per seed, with the full-seed versus inter-cluster-only split, which measures sensitivity to the intra-cluster order; (4) a second round that adds bitwise (word and bit-slice) seeds, a random-start and a coarse-random arm that separate the clustering from the coarse layout, and structure analyses of when seeding helps (Sections III-G and IV-A), with the research log in the appendix.

II. Background

A. Simulated annealing placement and RLPlace

VPR places by simulated annealing from an analytic initial placement [5], [6]. The anneal's moves are mostly random; its early phase can undo any structure the start carried. RLPlace improves the anneal rather than the start: a K-armed bandit selects among directed move types (simple moves and swaps, bounded centroid, median, critical-path moves), and checkpointing periodically saves the best placement so oscillation cannot throw it away [4]. Our experiment keeps the anneal as-is (VPR already ships RLPlace-style checkpointing, on by default) and changes the start instead; the checkpointing prediction – that save-best/restore extends the useful temperature range – is one of the hypotheses we test.

B. The seeds

The provenance hierarchy (Urd) is the module structure Odin III records on every netlist object; it needs no recovery. Urd has a single cluster only on sha and stereovision3, two on diffeq1 and diffeq2 (their one module plus a flat group), and one per populated module plus a flat group (3 to 23) on the other designs, so its signal is degenerate on the two one-cluster designs. The recovered clusters (Verdandi) are Loki III L2 partitions: Louvain, spectral, affinity propagation and k-means at several k, scored in the Loki III paper [7] against the known hierarchy. Agent default: Norns builds the Loki L0 graph.json from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. The two seed families differ exactly in the question's terms: provenance is the structure the designer wrote, clusters are the structure the flat netlist still shows.

III. Methodology

A. Seed construction

Each seed is a partition of the placed blocks. Its die-level arrangement comes from the quotient graph: contract each cluster to a node, weight edges by inter-cluster net cuts, lay the quotient out by a coarse force-directed solve, and scale it to the die. Inside a cluster, blocks take the seed's internal order, or a shuffle in the inter-cluster-only arm; the two arms differ only in that intra-cluster order, not in the coarse cluster layout.

B. Seed construction in detail

One die, one netlist. Every seed targets the same die: VPR's k6_frac_N10_frac_chain_mem32K_40nm architecture with six-input LUTs (K = 6). The netlist is the VTR-dialect BLIF that Odin III's front end writes for each design: read_verilog –nolatches –vtr-compat, proc, fsm, opt, flatten, opt -full, wreduce, hardmap, lower, abc –lut 6, stats, and write_blif –vtr –lut-k 6 –name-style provenance. Each configuration is one deterministic VPR run (–seed 1). The unseeded baseline run is the comparison point and also supplies the seed builders: its analytic initial placement (initial.place) gives every soft seed legal slots and its fine order, and its final.flat_place is the reference the retention metric uses.

Coarse layout. Every soft seed shares one die-level arrangement, the quotient-graph layout. Each cluster is contracted to one node and each pair of clusters is weighted by the number of nets cut between them; the weighted quotient graph is laid out by a force-directed solve (networkx.spring_layout, fixed seed), and the embedding is scaled onto the die's logic-slot bounding box. Each cluster then takes the free CLB slots nearest its laid-out point, nearest first. Clusters that exchange many nets therefore start close together, and the die fills in cluster-sized groups instead of a scatter.

Fine order. The soft arm has two variants that differ only inside a cluster. Full keeps the baseline analytic order: a cluster's blocks keep their relative analytic positions when they are assigned the cluster's slots. Inter shuffles the intra-cluster order under a fixed shuffle seed, so only the inter-cluster structure survives; because the coarse cluster layout is identical in the two variants, the pair isolates the fine order. The hard arm has no fine-order choice: it hands

VPR one PartitionRegion per cluster, sized from the full seed's slots and expanded until it holds the cluster's logic (and nudged to cover a memory or multiplier site the cluster needs), and VPR's placement orders the blocks inside each region.

Placement macros. Every seed honours the die's placement macros: I/O pads never move (their I/O sites are fixed by the architecture), mult_36 macros stay anchored on their tile footprint, and an adder carry chain moves as one vertical run in a single column, head at the top (VPR places a chain as a direct vertical macro). Chains are assigned slots before the nearest-first fill, so a chain is never split.

The seed families. The Urd seed is the provenance hierarchy made explicit: Odin III writes each block's HDL module into its netlist name, the seed builder reads that module from the BLIF, and each populated module becomes one cluster; blocks that carry no module name join one flat cluster. Urd is one cluster on sha and stereovision3, two on diffeq1 and diffeq2 (one module plus the flat group), and one per populated module plus the flat group on the other designs. The rationale is the designer's locality: modules are the units the designer grouped, and logic inside a module was written to communicate (shared words, state, control), so a module cluster concentrates the connectivity the placer is paid to shorten. That is the hypothesis the Urd arm tests.

The Verdandi seeds are Loki III's L2 clusterings of the flat netlist: Louvain, spectral, affinity propagation, and k-means at k, $2k$ and $4k$, with k the design's module count (two where the design has one module). Norns builds the L0 graph.json the clusterings read from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. A recovered cluster approximates the provenance grouping from connectivity alone, so comparing Urd with Verdandi isolates what the HDL structure adds over the graph.

Why a seed should beat a random start. A random placement scatters communicating logic across the die; the quotient layout puts each cluster's logic together and puts clusters that exchange nets next to each other, so the anneal starts from the structure it is meant to find and refines from there. The experiment tests that mechanism rather than assuming it, and it compares against VPR's analytic start, which already respects connectivity, not against a random placement. Two measurements report what the seed did: retention, the fraction of a seed's CLBs found inside its seed cluster's region at exit, has medians 0.68 (full) and 0.70 (inter) at VPR's own start temperature and 0.52–0.74 per family at twice the crossover; and the crossover sweep finds the largest tested temperature scale at which the seed still wins, the point where its contribution stops surviving the anneal.

What the HDL constructs say about a good start. HDL modules, buses, datapath words and finite-state machines group logic that communicates: a bus is one word fanned out to every consumer, a datapath chain is a dataflow line, and an FSM's next-state logic and registers exchange signals every cycle. Provenance preserves that grouping in the netlist, so a seed built from it starts the placer with the designer's locality already on the die; L2 clusters try to rediscover it from connectivity alone, and this experiment measures how well that substitute places, not how closely the clusters match the modules. That is why Urd and Verdandi are the arms, and why the paper asks whether recovered structure is interchangeable with provenance.

C. Injection

Seeds enter VPR two ways, both already in the pinned build (v8.0.0+18416): soft, as an initial placement file (–read_initial_place_file) that the anneal is free to move away from; and hard, as floorplanning constraints (–read_vpr_constraints) with one PartitionRegion per cluster, which the anneal cannot violate at any temperature. The hard arm constrains the regions only, so the anneal chooses the fine placement inside each one (one hard variant, no full/inter split). –write_initial_place_file dumps VPR's own analytic start for the baseline study. VPR's placement macros are respected: I/O pads and the multiplier tiles stay anchored, and carry chains move as one vertical run. The seeds are built from an Odin III provenance BLIF and Loki III's clusterings of the same netlist, arranged by a cut-weighted quotient-graph layout with networkx (spring layout, fixed seed). Building the seed files also needs VPR's unseeded baseline placement for legal slots, so each design's baseline run precedes its seeds.

D. The crossover sweep

The calibration loop follows the data, in order. First, run every seed and keep the best final placement P^* (and which seed made it). Second, cold-start a fresh anneal from P^* and sweep the initial temperature upward; the crossover is the largest tested scale at which the cold start still routes shorter than the unseeded run at the same scale, a lower bound on the true fall point when the sweep stops at its cap. The sweep is run once per design, so S is design-level and identical across that design's seeds. The temperature is reported as a multiple S of VPR's automatic start-temperature estimate. Third, run every seed at twice the design's crossover against an unseeded run at the same temperature: whether the seed still wins there tests how much of its advantage survives a hot restart. The checkpointing arm repeats the sweeps with VPR's checkpointing disabled; per RLPlace, the crossover should move hotter when it is on.

E. Metrics and data

Per run: seed retention (fraction of cluster CLBs still inside their seed cluster's region at exit; mean displacement from the seed coordinate), routed wirelength, critical-path

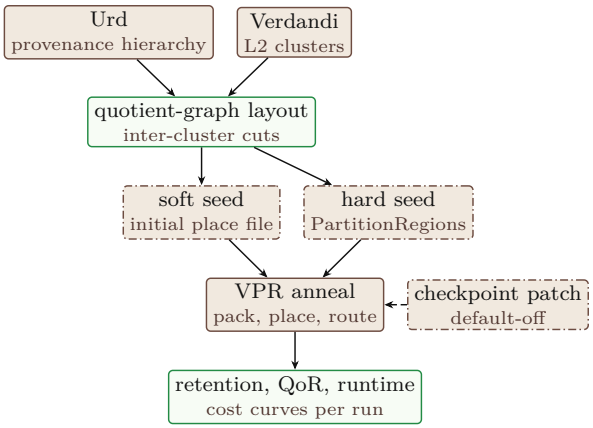


Fig. 1. The Norns III experiment. Seeds from provenance and recovered clusters, arranged by a quotient-graph layout, enter VPR softly (initial placement) or hard (constraints). The crossover sweep cold-starts from the best placement and raises the initial temperature until the seed's advantage dies or the sweep reaches its cap; the checkpointing arm (VPR's stock flag, on vs off) tests whether save-best/restore extends the range.

delay, minimum channel width, placement runtime, and the per-temperature cost curve. Designs are the 15 VTR benchmarks that fit the development machine, smallest first, one run at a time under the heavy lock. Every number in this paper traces to a kept record; large data keeps its scripts.

F. Benchmarks

Table I reports each design's construction and its post-technology-mapping size. The module count is the front end's, while the CLB count and the grid are VPR's own numbers for the packed netlist, so the table describes the designs as the experiment actually places them.

The 15 designs span two orders of magnitude after technology mapping: from 13 CLBs (ch_intrinsics) to 1956 (stereovision2), 4843 CLBs in total. The mean is 322.9 CLBs and the median 125; the mean sits well above the median because three stereovision designs carry most of the logic. VPR auto-sizes a square grid to the packed netlist, with a mean side of 27.2 tiles (median 20) and a mean of 1076.4 grid tiles (median 400). Grid size follows the largest resource demand, not the CLB count: I/O-heavy and memory-heavy designs such as mkDelayWorker32B get a large die for few CLBs.

G. Round two: bitwise seeds and structure analyses

Round one seeded VPR with module-level clusters. Datapath logic is bit-level: a 32-bit register or adder is a regular stack of identical bit cells, and the carry chain already shows that VPR places such stacks as ordered vertical runs. Round two therefore adds a bit-level seeding aspect that keeps the module hierarchy at the coarse level (Section A records where the idea came from).

a) Word membership from Odin III data, not names.: The obvious source, the provenance BLIF's block name module/$nN@file:line, is too coarse: the line is the HDL statement, so one always block is one group (the largest (module, file:line) group in any design holds up to 5248 blocks, and sha's holds 1739 of its 2506 blocks; 370 of the 790 statement lines that hold word blocks hold more than one word, up to 320). The finer records exist in the Odin III session that writes the BLIF, so the seed builder reruns the identical script, checks that the BLIF it writes equals the kept one byte for byte (all 15 designs match), and reads the records Loki III's labeller reads: the HDL name bus[i] of a cell's output net and the source spans of each cell, matched to the word-level cells that exist before technology mapping. A final cell joins its BLIF block through the net id the BLIF name carries ($n<id>), so the join is by identifier, not by string (1204 of the 106262 Odin III final cells, 1.1%, found no BLIF block and are left unlabelled, from none to 22.2% per design, the worst being mkDelayWorker32B with 942 of 4235; the writer legalises dotted names, handled explicitly). A block's word is the first rule that applies: (bus) an output net named bus[i] puts it in word bus at bit i, which holds for every flip-flop that keeps a register name; (cell) exactly one word-level cell's spans lie inside the block's spans, which is true for adders, multipliers, memories and single-statement LUTs; otherwise (none) the block has no word. Bit order inside an adder word is the carry-chain position. ABC's cone provenance merges the spans of a LUT that mixes words, and its edge budget makes some runs share one record, so 49% of the LUTs get a word against 100% of the flip-flops and 99% of the hard cells. Over all designs, 80.6% of the blocks are in a word (per-design median 66.7%); 5190 words in all.

b) Seed families.: A CLB joins a word cluster when at least one third of its atoms carry a word and that word is the plurality; otherwise it keeps its Urd module cluster. Four families follow, each entering VPR as a soft initial placement at VPR's own start temperature (scale 1). words: the word CLB groups are the clusters, arranged by the round-one quotient-graph layout. slices: the same clusters, but each word is laid out as a vertical run in bit order (bit 0 on top, the carry-chain direction), runs of equal length aligned to the same rows, so that bit i of connected words shares a row; true carry chains are untouched. hier: Urd module clusters at the coarse level and, inside each module, the words as bit-sliced runs. words_partial: only the word CLBs are seeded, every other CLB keeps VPR's analytic position (and every carry chain with an unseeded CLB stays whole). The words family also has the inter-cluster-only variant. Pads and multipliers stay anchored as in round one.

c) Random and coarse-random arms (RQ-N4).: random is a random legal start: every logic CLB takes a random slot and carry chains a random window, fed as a soft seed at scale 1. The coarse-random arm keeps the

TABLE I
Benchmark construction and post-technology-mapping size. All fifteen designs are from the VTR 8.0 benchmark set (vtr_flow/benchmarks/verilog). Modules is the front end's module count, CLBs the clb blocks VPR packs, Memory and Mult the memory and mult_36 blocks, and Grid the auto-sized device grid with its tile count in parentheses. Sizes are read from each design's baseline run log and kept in data/benchmark-sizes.json.

Design	Suite	Modules	CLBs	Memory	Mult	Grid (tiles)
ch_intrinsics	VTR 8.0	2	13	1	0	10×10 (100)
stereovision3	VTR 8.0	1	15	0	0	7×7 (49)
diffeq2	VTR 8.0	1	22	0	5	16×16 (256)
diffeq1	VTR 8.0	1	37	0	5	16×16 (256)
mkPktMerge	VTR 8.0	7	21	15	0	28×28 (784)
raygentop	VTR 8.0	15	143	1	6	19×19 (361)
sha	VTR 8.0	1	142	0	0	17×17 (289)
boundtop	VTR 8.0	13	54	1	0	15×15 (225)
mkSMAdapter4B	VTR 8.0	8	125	5	0	20×20 (400)
or1200	VTR 8.0	16	252	2	1	27×27 (729)
stereovision0	VTR 8.0	25	707	0	0	33×33 (1089)
stereovision1	VTR 8.0	16	788	0	43	40×40 (1600)
mkDelayWorker32B	VTR 8.0	16	74	43	0	50×50 (2500)
stereovision2	VTR 8.0	24	1956	0	192	82×82 (6724)
blob_merge	VTR 8.0	2	494	0	0	28×28 (784)

clusters and their within-cluster order but permutes the cluster positions on the die (urd_shuf for Urd, and the *_shuf twin of the best new family).

d) Structure analyses (no VPR runs).: RQ-N1 measures, on each design's unseeded baseline, how module- and word-coherent three placements are: VPR's analytic start, its final placement, and a random null (logic-CLB coordinates permuted, 20 samples). A logic CLB carries the plurality label of its atoms (module: the Urd module; word: the word with most atoms in the CLB; a CLB without one is unlabelled). Coherence is the NMI between a $g \times g$ grid partition of the labelled CLBs ($g = \max(2, \lceil\sqrt{\#\text{labels}}\rceil)$, the grid spanning the labelled CLBs' bounding box) and the labels. The compactness is a separate metric that does not use the grid: the CLB-weighted mean over labels with at least two CLBs of the label's CLBs divided by the area, in tiles, of the label's bounding box. A labelling is scored only with at least four labelled CLBs and two labels, and a design is counted against the null only when the null's standard deviation is positive (counted separately for the two metrics). A design with fewer than two real Urd modules (diffeq1 and diffeq2, which have one module plus the artificial flat group, and sha and stereovision3, which have none) has no module row. Label sets with fewer than 20 labelled CLBs are flagged in the results. RQ-N2 correlates the seed gain with three structure measures of the netlist graph: Louvain modularity Q; the z-score of Q against a degree-preserving null (double edge swaps, 10 per edge, 20 samples); and the inverse Rent exponent $-p$, with p fitted from the terminals of BFS balls of geometric sizes (a lower exponent is a more local design). Spearman's ρ and p-value are reported with $n = 15$, which is small, and again as partial correlations with the edge count regressed out, because graph size drives both $z(Q)$ and the gain.

e) Hypotheses and decision rules.: Paired comparisons are per design (variant over reference), reported as a geomean ratio, the number of designs where the variant is better and a two-sided exact sign test. H7a: words beats Urd in soft wirelength (supported if the geomean is below 1 and $p < 0.05$, not supported if the geomean is at least 1, else inconclusive). H7b: slices beats words on wirelength and channel width (supported if both geomeans are below 1 and the wirelength sign test has $p < 0.05$). H7c: hier has the lowest geomean wirelength of all eleven soft families. H7d: words_partial is better than the full word seed (supported if the geomean ratio is below 1 and the wirelength sign test has $p < 0.05$, not supported if it is above 1 with $p < 0.05$, else inconclusive; no equivalence margin was fixed, so "at least as good" is not testable here). H1: the baseline's final placement is module- and word-coherent (supported if, for both labellings, at least two thirds of the designs lie more than two null standard deviations above the null in NMI and at least two thirds in compactness, the two counted separately; not supported if fewer than a third do for both labellings and both metrics). H2: structure predicts gain (supported if at least two of the three measures correlate positively with the best round-two gain at $p < 0.05$; not supported if no correlation of either sign reaches $p < 0.05$). H4: random start $>$ permuted clusters $>$ quotient layout $\geq$ analytic start in wirelength ratio, for Urd and for the best new family.

f) Which rules predate the data.: These rules were not registered anywhere outside the repository, so their order is stated from the commit times of that repository. The decision-rule code for H7a to H7d and H4 was committed at 15:21 on 2026-10-24 and H1's rule (with the compactness criterion) at 15:24, before the first round-two VPR data (15:38). Some of those rules were later found weak and changed; the appendix lists the changes. After the first data: tie handling in the sign tests (ties left

out) came with the first data commit; the methodology text that states the rules was written at 17:31, after nine designs; the best family was frozen at 13 designs after their results were seen (so the hard, crossover and permuted arms of the best family use a choice made on data); the H2 rule was briefly changed and reverted, and the H7d rule was replaced after review. The next round fixes every decision rule, and the equivalence margin of H7d, in a commit before any run.

H. Round three: packing, I/O words and placement proximity

Round two found every soft seed within a few percent of the baseline. Round three asks whether the packing step, which runs before placement and decides which atoms share a CLB, is where the HDL structure is lost, builds a packer that keeps it, and then asks what the annealed placement itself says about provenance. The two questions that started it are quoted in the research log (Appendix A).

a) The Norns packer.: VPR's packer is AAPack, the greedy clusterer in vpr/src/pack (not T-VPack); in VTR's default analytical flow the same clusterer runs as APPack, with candidate gains attenuated by distance in the global placement. The Norns packer is a copy of the changed files of that clusterer (VTR commit 3c9a4d23b2, MIT licence kept) with one added term: an atom carries a group id read from a file (–norns_affinity); a cluster's dominant group is the plurality group of its atoms so far; a candidate molecule gets $+\beta$ (–norns_affinity_weight) for every atom in that group, and up to 30 unclustered members of the group are proposed as candidates, because the bits of a word are parallel and no net links them. The stock per-atom gain lies in $[0, 1]$, so β is in the units of one connected atom's gain. Legality stays with AAPack's legalizer. Erratum found in round four (Section III-I): the affinity files of this round omitted every adder (carry-chain) atom, because the builder named adders by their first pin; the purity and wirelength results below describe a packer that did not see carry chains. The change is 8 files, 536 added lines and 6 removed. Control gate: with the option absent and with $\beta = 0$, the packer must write a .net byte-identical to the stock vpr's in –pack and in the analytical flow on all 15 designs, before any experiment run: 60 comparisons over 15 designs, all byte-identical; the same $\beta = 0$ check for the module and word+io affinity files (–pack, seed 1): 30 of 30 byte-identical.

b) Limits of the proposal step.: The step hands up to 30 unclustered members of a cluster's dominant group to the candidate queue, scanning the group in netlist (atom id) order. Its counter counts the molecules handed over, not the molecules the queue accepted, so the cap of 30 can be spent on molecules the queue ignores (already queued, or beyond APPack's distance threshold), and the scan always starts at the lowest atom ids of the group: which members are tried first is biased by netlist order. The behaviour is unchanged in this paper (Stage-4 results stand); it was measured with a log-only counter added to the packer, which leaves the packing unchanged (45 of 45 .net files byte-identical to the previous binary's, 45 of 45 flat placements identical to the Stage-4 run; pack-only style runs of the headline configurations, seed 1, no routing). Share of handed-over molecules that entered the queue, median over designs (module / word / word+io): 48.8% / 46.4% / 47.6% (pooled over all calls: 47.4% / 53.1% / 52.6%); proposal calls (one per cluster) that queued nothing: 13.1% / 54.6% / 49.2% (Table XV). The fix (count only queued molecules, return a bool from the queue step, a scan order not tied to atom ids) is queued for round 4.

c) RQ-N8, dilution (no VPR).: On the stock packings of round one, per design: purity (plurality share of the labelled atoms of a logic CLB, atom-weighted), and the split factor (CLBs a word spans over the fewest CLBs that could hold it, 1.0 being best), at word and module level, against a null that assigns the same atoms at random to CLBs of the same sizes (50 samples). Round two's word coverage loss is split into the front end (atoms with no word) and packing (word atoms not in a CLB where their word is the plurality word under the 1/3 rule).

d) RQ-N10, I/O words.: A top-level bus name[i] is an io-word; its identity is propagated inward over LUT and flip-flop atoms, two hops past the atom next to the pad, through nets of fanout at most 16, giving an atom that has no round-two word the io-word it reaches from exactly one bit of exactly one io-word.

e) RQ-N9, packing experiments.: Arms: the stock packer, and the Norns packer with affinity source module, word or word+io (word, else io-word) at three values of β each, with the analytic start or with the round-two words_partial seed rebuilt on that run's own packing, three VPR seeds (1 to 3) per configuration. The grid was chosen on a pilot of the four smallest designs that reads only purity and CLB count (never wirelength): β_m is the largest value in $\{0.05, 0.1, 0.2, 0.5, 1, 2, 4\}$ whose median CLB-count growth is at most 5%, and the grid is its neighbours (module: $\beta_m = 4$, grid 1, 2, 4; word: $\beta_m = 4$, grid 1, 2, 4; word+io: $\beta_m = 4$, grid 1, 2, 4). The pilot did not discriminate between the betas. The median logic-CLB growth over the four pilot designs was negative at every β of the list (largest value -3.8%), so the 5% growth rule passed all of them and took the end of the list, and the purity gain of the word sources was the same at $\beta = 2$ and $\beta = 4$ (module: zero throughout). The grid $\{1, 2, 4\}$ is therefore not informed by the pilot, and a larger gain scale is untested. Each claim is reported on all 15 designs and without the four pilot designs; a status that differs between the two is called inconclusive (pilot-sensitive).

f) RQ-N11, proximity.: On the stock-packing, analytic-start runs, logic CLB pairs by relation (same word, same module, same statement, net-connected, same io-word) against the direct net count, which is stratified

out: the stratified mean distance difference of related over unrelated pairs, as a fraction of the unrelated distance (negative is closer). A stratum counts only with at least 5 related and 30 unrelated pairs. The status is computed under the protocol text (4fb8b60: no design excluded, the sign rule needs at least 12 of the 15 designs, so a relation defined for fewer designs cannot be supported), which is the primary status, and the following variants are reported beside it, labelled. (i) The sign rule scaled to $\lceil 12n/15 \rceil$ designs, which the first analysis code used; the protocol does not register it. (ii) The amendment d805789: a design is left out of a relation when 95% or more of the sampled pairs are related (applied to the module relation and to the word relation, per design on the mean over the three seeds, with the scaled and the literal count), and the label permutations are 200 per design and seed. The primary permutation test is the protocol's 1000 label permutations pooled over designs; it is reported, not used in a status. The entry of 2026-10-25 in docs/plan/round3.md records these deviations.

g) Decision rules and their provenance.: Every hypothesis, metric, arm and numeric rule (including the margins ±2% and 1% for wirelength, 5 percentage points of purity, 5% CLB growth) is in docs/plan/round3.md, committed as 4fb8b60 on 2026-10-24 21:26 (UTC-4), before any round-three measurement or VPR run. Two later dated entries exist and the earlier entries are not edited. The entry d805789 (2026-10-24) is titled a clarification, but it is an amendment: besides definitions it changed two H11 rules (the label permutations, 1000 pooled over designs, became 200 per design and seed; a new exclusion of designs with 95% or more related pairs). It was committed at 22:27:05, before any Stage-4 result was analysed, but the Stage-4 runs of ch_intrinsics (60 runs, finished 22:26:38) and 40 of the 60 of stereovision3 had already finished. The entry of 2026-10-25 (the fix round after review) records what the analysis code did beyond the protocol text. Where the code deviated, the primary status in this paper is the one under the text of 4fb8b60 and the deviating variant is reported alongside, labelled.

I. Round four: method

Protocol docs/plan/round4.md, committed as fbddfa0 on 2026-10-25 12:51 (UTC-4), before any round-four measurement or VPR run (the packer fixes and the Stage-1 gate runs followed it; a later dated entry fixes construction details and amends N16 items 5 and 6, before any N16 data, keeping the old text; a fix-round entry records corrections and disclosures; no hypothesis rule changed). Small test runs on ch_intrinsics (options parse, census writes) preceded the gates; they were deleted and no result uses them. Designs, flow and the Norns headline (source word, $\beta = 4$) are those of round three.

a) Packer fixes and gates.: The proposal step now counts molecules that entered the candidate queue (–norns_proposal_count queued); attempted is the round-three behaviour. A log-only census (–norns_pack_census) writes, per molecule added to a cluster, the candidate source, the gain terms VPR held, the pack-time criticality of each input pin, the molecules offered to the candidate queue and the candidates the legalizer refused. Gates (15 designs, seed 1, byte comparison of the .net): (a) stock against the Norns binary with the option absent and beta 0, 30 of 30 identical; (b) census on against off, –pack and the analytical flow, stock and headline arm, 60 of 60; (c) attempted counting against the round-three binary on the 45 headline configs (.net and flat placement), 90 of 90. No experiment ran before all passed. Fix round: (b′) census on against off for the round-three attempted arm (word, $\beta = 4$, –pack, seed 1), 15 of 15 identical; the flat-placement compare now skips only the # Created: header line. The gate scratch outputs had been deleted, so (b) and (c) were not repeated; the tightened compare was run on the 30 pairs of existing output (the census run and the routed arm run of one packing, seed 1): 15 identical as is, 30 when the stock vpr's version line is also skipped, 30 with the old compare.

b) Affinity join.: Round three's affinity files matched few entries for mkDelayWorker32B (38 of 1620 word entries). The cause was not a name form: the file listed buffer LUTs that VPR absorbs, dead logic that VPR sweeps, and (a bug found in this round) it named every adder after its first pin instead of its cout net, so adder atoms were never in a round-three file. The round-four builder drops absorbed buffers and swept blocks and names adders as VPR does; every entry of every file now matches an atom (Table II). The round-four files were final at 13:08 (join check 13:12), after the first dated entry described them as "the same atoms as round three's minus buffers"; the entry of the fix round records the correction. Because the adders are now labelled, the round-four files change the packing of 33 of 45 (design, source) pairs against round three's (information rows of the gate).

c) Census classes.: Every pair of LUT, FF and adder atoms in one logic CLB falls in the first class that applies: series-critical (a local net, fanout $\leq$ 32, from one to the other, with pack-time sink criticality ≥ 0.5), series-noncritical (the same below 0.5), shared-input (sinks of one common local net), orthogonal (same round-two word, or sinks of one common broadcast net), filler. Fifty null packings (same CLB sizes, random assignment) give the chance level. Post-route criticality comes from the Odin III critical-path attribution (V1.1.1) of the routed runs; because the Norns BLIFs already carry provenance names the tool's two-BLIF input does not exist for them, so scripts/timing4.py builds the name map from the Norns BLIF and calls the tool's own functions (a labelled workaround; the odin3 checkout is not edited). Coverage: the share of critical-path nodes the attribution matches to a provenance name is at least 0.58 in every one of the 264 routed runs, and 1.00 in every run when port nodes (which

have no HDL line) are excluded. The routed report holds one path per endpoint (at most 1000), so near-critical counts are per reported path, not per start and end pair.

d) Depth estimate (H15).: The HDL statement of a span is the statement that starts on the span's source line or, if none does, the nearest statement start within 12 lines above (a rule written in scripts/r4depth.py before any data, not stated in the protocol text).

e) Stage-by-bit seed.: Stage = register-to-register level of the atom graph (back edges removed by a DFS), bit = round-two word bit, control = CLBs holding the driver of a net of fanout > 32. Moved CLBs (no carry-chain CLBs) permute among their own analytic slots, each taking the nearest free slot to (stage, bit), control in the centre; all other CLBs keep their position.

J. Round five: method

a) Question N Q1.6.: On designs with a known optimal clustering, how far are the VPR packer and the Norns HDL-aware packer from it, and how does the gap grow with size? The idea follows Cong, Romesis and Xie [8], who built placement problems with known optima to measure how far placers are from them.

b) What "optimal" means here.: A clustering optimum by construction, not a proven placement optimum. A block is 10 six-input functions, each feeding a flip-flop (10 logic elements of the k6_frac_N10 architecture, so exactly one CLB), with dense internal connections and at most two external inputs per function. The CLB count B is an exact lower bound and is reached by the block partition. The block partition's cut is exact for that partition but is an upper bound on the minimum cut: at the netlist level a local search of improving pair swaps found a slack of 0.0% to 6.8% (median 0.0%) on the 72 netlist designs; the search was capped at 8 rounds and converged within the cap on 72 of 72 designs (at most 3 rounds), so the slack is a lower bound on the non-minimality of the block cut. The random-restart search (protocol: 20000 restarts; run: 500 at $B = 4$ and 300 at $B = 8$, 16 designs) found a smaller cut on 0 of them, but its best cuts at $B = 8$ (79 to 86) are far above the block cuts (30 to 56), so it is weak evidence that the block cut is minimal. The "known" arm hands the block partition to the same placer and router, so its wirelength, channel width and critical-path delay are what VPR achieves from a good clustering, not a bound; ratios below one are possible.

c) Designs.: Nine sizes ($B = 4$ to 1024), four topologies (chain, mesh, tree, random sparse), two styles (dp: a datapath slice whose HDL has the vector r[9:0]; ctl: scalar registers, random logic with no vector), generated deterministically from the seed 20261026. Two levels: NL, the netlist written directly as BLIF (no Odin III, no ABC), and HDL, the same specifications as Verilog through Odin III. Arms: stock VPR; norns (HDL level only: the round-four headline packer, affinity source word, $\beta = 4$, which needs provenance); known (the Norns binary with the block membership as affinity groups, $\beta = 100$). Three VPR seeds each. Synthesis keeps the construction in 63 of 72 HDL designs under the amended label rule (original, literal rule: 32); the netlist check passes on 37 of 72 designs under the literal zero-violation rule, 72 of 72 under the slack rule of entry (c) (chosen after the data) and 63 of 72 at a 5% slack bound. H19 to H24 use the 63 designs of the amended rule; H25 to H27 the 72 designs of entry (c). Excluded by the rule fixed in advance: r5_chain_ctl_1024, r5_chain_ctl_512, r5_chain_ctl_64, r5_mesh_ctl_1024, r5_mesh_ctl_256, r5_mesh_ctl_512, r5_tree_ctl_1024, r5_tree_ctl_32, r5_tree_ctl_512. Rules, thresholds and statuses are in the protocol (docs/plan/round5.md, SHA 7886f49) and six dated entries; the paper repeats none of them.

IV. Results

The 15 completed designs give the ratios of Table IV. Soft seeds stay within a few percent of the unseeded baseline in routed wirelength (1.009 for the full seed, 1.015 for the inter-cluster-only arm, both pooled over all seven seed families) while cutting the minimum channel width to 0.973 and costing 1.087 in placement runtime (both also pooled). The hard arm is worse on average (geomean 1.088 over the 104 runs that completed; 14 of those still beat the baseline, and stereovision3's hard geomean is 0.906), but not uniformly: one blob_merge k-means$_{2k}$ run is missing from the arm because VPR's initial placement could not fit 9 CLBs inside its regions (the constraints are too tight, not a packing failure). Fixed regions may block the anneal's fine moves across cluster borders – a hypothesis the aggregate numbers are consistent with. No single seed family dominates: Urd is the worst full seed in soft wirelength (1.029) and the best hard seed (1.062 against 1.075-1.120 for the recovered families). Retention at scale 1 is 0.68 of a seed's cluster CLBs (full) and 0.70 (inter).

A. Round two results

RQ-N7, bitwise seeds (15 designs, one VPR seed each). H7a is inconclusive: words over Urd is 0.986 in wirelength, better on 10 of 15 designs (sign test $p = 0.302$). H7b is inconclusive: slices over words is 0.982 in wirelength (better on 8 of 15, $p = 0.791$) and 1.033 in channel width (better on 6 of 15). H7c is not supported: hier ranks 8 of 11 families at 1.014; the lowest is words_partial at 0.994. H7d is inconclusive (no evidence of a difference): words_partial over words is 0.980 (better on 9 of 15, $p = 0.607$). Ties (identical results) are left out of the sign tests: 0 for H7a, 1 for H7b and 0 for H7d. A family with nothing to rearrange reproduces its reference exactly (a design whose words are all single-CLB, or that has one module): slices equals words on 1 design(s), hier equals Urd on 1 and words_partial equals words on 0. Seed audit. Rebuilding the seven round-two seed families

TABLE II

Affinity join. Forward: affinity-file entries that match a VPR atom, round three (r3) and round four (r4), per source. Reverse (r4 files): VPR LUT, FF and adder atoms in logic CLBs that carry a label (by their own name) and have an entry in the file; the last column is the same for the round-three module file's adder atoms (it had none).

	module		word		word+io		reverse, r4 file			r3
design	r3	r4	r3	r4	r3	r4	module	word	word+io	adders
ch_intrinsics	144/144	144/144	81/113	81/81	113/177	113/113	144/144	81/81	113/113	0/0
stereovision3	261/261	289/289	162/162	190/190	187/189	215/215	289/289	190/190	215/215	0/28
diffeq2	198/230	330/330	160/160	292/292	260/260	392/392	330/330	292/292	392/392	0/132
diffeq1	391/423	523/523	257/257	389/389	353/353	485/485	523/523	389/389	485/485	0/132
mkPktMerge	191/191	214/214	180/180	203/203	180/333	203/203	214/214	203/203	203/203	0/23
raygentop	1836/1931	2255/2255	1186/1234	1603/1603	1575/1634	1992/1992	2255/2255	1603/1603	1992/1992	0/419
sha	2147/2275	2456/2456	925/925	1234/1234	1507/1507	1816/1816	2456/2456	1234/1234	1816/1816	0/309
boundtop	687/762	748/748	555/620	616/616	674/748	735/735	748/748	616/616	735/735	0/61
mkSMAdapter4B	1581/1638	1892/1892	673/787	978/978	970/1094	1275/1275	1892/1892	978/978	1275/1275	0/311
or1200	2719/2783	3233/3233	1317/1350	1751/1751	1870/1947	2304/2304	3233/3233	1751/1751	2304/2304	0/514
stereovision0	18148/18148	20836/20836	13666/13667	16354/16354	14213/14244	16901/16901	20836/20836	16354/16354	16901/16901	0/2688
stereovision1	16536/16632	18710/18710	13472/13568	15622/15622	13908/14017	16058/16058	18710/18710	15622/15622	16058/16058	0/2174
mkDelayWorker32B	469/1566	536/536	38/1620	104/104	38/1643	104/104	536/536	104/104	104/104	0/67
stereovision2	23877/32721	35886/35886	22273/22274	34282/34282	22797/22811	34806/34806	35886/35886	34282/34282	34806/34806	0/12069
blob_merge	6490/6500	10011/10011	3507/3507	7004/7004	3677/3716	7174/7174	10011/10011	7004/7004	7174/7174	0/3521

TABLE III

Proposal step, median over designs (census runs, seed 1): fraction of offered molecules that entered the queue and of proposal calls that queued nothing.

arm	queued / offered	calls with 0 queued
round-three counting	46.6%	58.3%
fixed counting	48.2%	55.6%

TABLE IV

Geomean quality ratios vs the unseeded baseline, per seed family, for the soft initial-placement arm; the inter-cluster-only row shuffles the order inside each cluster and pools all seven seed families, and the k-means row pools k, $2k$ and $4k$. Below 1.0 is better. The hard-constraint arm is discussed in the text.

Seed	Wirelength	Delay	Chan. width	Runtime
Urd (hierarchy)	1.029	1.004	0.958	1.068
Louvain	1.008	0.982	0.980	1.091
Spectral	1.006	0.993	0.968	1.098
Affinity propagation	1.002	1.003	0.982	1.086
k-means (k, $2k$, $4k$)	1.006	0.985	0.975	1.088
Inter-cluster only	1.015	0.981	0.987	1.102

TABLE V

The crossover sweep, per seed family: the largest tested scale S of VPR's automatic start-temperature estimate at which the cold start from P^* (the best placement over all seeds) still routes shorter than an unseeded run at the same S, and the seed's median retention at the hot step ($2S$). S is design-level (one sweep per design), so it is identical across families; where the sweep stopped at its cap, S is a lower bound.

Seed	Crossover S	Retention
Urd (hierarchy)	16×	0.744
Louvain	16×	0.520
Spectral	16×	0.635
Affinity propagation	16×	0.564
k-means (k, $2k$, $4k$)	16×	0.667

TABLE VI

Round two: geomean ratios vs the unseeded baseline at scale 1 for the soft arms (Urd from round one for reference). The last column is the number of designs where the routed wirelength is below the baseline. WL is routed wirelength, Chan. the minimum channel width, Time the placement runtime, T_0 the anneal's start temperature that VPR derives from the starting placement. Below 1.0 is better (except T_0, a diagnostic).

Seed	WL	Delay	Chan.	Time	T_0	Better
Urd (round 1)	1.029	1.004	0.958	1.068	3.23	3/15
Words	1.015	0.980	0.964	1.254	6.24	5/15
Words, inter only	1.017	0.984	0.970	1.264	7.35	6/15
Slices	0.996	0.986	0.995	1.268	8.07	9/15
Hier	1.014	1.002	0.982	1.247	4.52	6/15
Words, partial	0.994	0.980	0.962	1.257	4.63	7/15
Random start	0.997	0.978	1.012	1.275	9.24	8/15
Urd, permuted	1.020	0.992	0.972	1.227	3.09	4/15
Words, partial, permuted	0.980	0.972	0.988	1.179	5.19	7/15

from the kept word membership with the committed builders gives seed files byte-identical to the ones VPR ran from on all 15 designs; so every slices run used the column-major bit-order fallback of the committed code, and that fallback changes the slices seed on 6 designs (ch_intrinsics, diffeq1, mkPktMerge, raygentop, stereovision0, stereovision2). The designs where slices equals words as a seed file (mkDelayWorker32B) and where hier equals Urd (mkDelayWorker32B) are equal because there is nothing to rearrange, not because a code path was skipped. On mkDelayWorker32B only 9 CLBs carry a word label, in 7 words, and no word has two CLBs off the carry chains.

words_partial places only the word CLBs, so how much of each design it moves matters for H7d (Table XXV). Of the logic and memory CLBs the seed moves a median of 79% per design (6 of 15 designs at 90% or more, down to 15% on mkDelayWorker32B); counting every placed block, pads and multipliers included, which are anchored in every arm, the median is only 18% (for example 11 of 243 on ch_intrinsics). Where the seed covers nearly all logic CLBs, words_partial differs little from words, so H7d compares two similar placements; where it covers

little (for example mkDelayWorker32B), most of the seed is VPR's own analytic start. H7d is therefore a weak test of "seeding fewer CLBs" on the designs where the share is high. The best new family by geomean wirelength is words_partial (chosen on the 13 designs whose soft arms were complete when the hard and crossover arms were queued), so the coarse-random, hard and crossover arms below use it. Its geomean of 0.994 is a selected figure: the family was the best of four on those 13 designs and is reported on them (and on the 2 added later), so it is biased low, and it lies inside the band of the other arms. Its hard arm (one region per cluster, scale 1) is 1.141 in wirelength over 12 designs, better than the baseline on 2; round one's hard arm pooled to 1.088 (all families, all designs). That figure is over the 12 survivors only, and the 3 designs that failed are among the largest, so survivors flatter it. Round one's Urd hard arm over the same 12 designs is 1.042 (better on 3): on equal designs the bitwise family's hard arm is worse than Urd's. The hard arm failed in VPR's initial placement on 3 designs (stereovision0, stereovision1, stereovision2: no free sites for the constrained blocks, the same failure round one had on blob_merge) and those are left out. The builder's region check counts a box as large enough when it holds as many logic sites as the cluster's own CLBs; it does not count the unseeded CLBs that this family leaves at their analytic positions, which can sit inside a box, so these failures may be builder-side rather than a property of hard regions. Its crossover sweep, started from its own scale-1 placement, has median crossover scale 16 against 16 for round one's best-of-seven P^* over the 13 of 15 designs where it won at some scale (at least round one's on 11 of 13); at twice the crossover the retention median is 0.20 and the seed beats an unseeded run at the same temperature on 5 of 13 designs. stereovision3 and mkPktMerge had no winning scale at all and are left out of those medians and counts (round one won up to 256× on stereovision3 and 16× on mkPktMerge). Counting them as failures, the family is at least round one's on 11 of 15 designs, and the median over all 15 (a failure taken as below the lowest scale) is 16 against 16.

RQ-N4, clustering versus coarse layout. H4 is not supported. For Urd the wirelength ratios are random start 0.997, permuted cluster positions 1.020, quotient layout 1.029; for words_partial they are 0.997, 0.980 and 0.994. The analytic baseline is 1 by definition, so a random start below 1 means the anneal from a random start ended with shorter wire than from VPR's analytic start; the gap between the permuted and the quotient arm is what the coarse layout is worth beyond the clustering itself. Permuting the cluster positions changes the wirelength by -0.9% for Urd and -1.4% for words_partial relative to the quotient layout (negative is shorter).

RQ-N1, does VPR rediscover the hierarchy? H1 is inconclusive. Table VII gives medians over designs: the final placement is above the random null in NMI on 8 of 11 designs for modules and 9 of 15 for words, and in compactness on 10 of 11 and 14 of 15 (more than two null standard deviations). Modules are counted over 11 designs, not 15 or 13: diffeq1 and diffeq2 have a single module, whose second label is the artificial flat group, and sha and stereovision3 have none. Label sets with fewer than 20 labelled CLBs carry little information (module labels on ch_intrinsics; word labels on ch_intrinsics, stereovision3, mkDelayWorker32B). Final and analytic placements are close, and the final NMI is at least the analytic one on 5 of 11 (modules) and 7 of 15 (words) designs. With many small labels the NMI of a random placement is itself high (word null 0.66), so compactness is the cleaner signal.

TABLE VII
RQ-N1: medians over designs of the grid-partition NMI and label compactness of the analytic start, VPR's final placement and a random null (20 permutations). Designs with fewer than two real modules have no module label and are left out of that row.

Labels	n	NMI (start/final/null)	Comp. (start/final/null)
Module	11	0.26/0.31/0.12	0.20/0.26/0.09
Word	15	0.72/0.73/0.66	0.33/0.32/0.08

TABLE VIII
RQ-N2: Spearman ρ (p) between per-design structure and seed gain (1 - wirelength ratio): Urd round one, best of seven round-one families, best new family. The inverse Rent exponent is $-p$.

Gain	Q	$z(Q)$	$-p_{\mathrm{Rent}}$
Urd, round one	-0.17 (0.533)	-0.44 (0.098)	0.12 (0.657)
Best round one	-0.09 (0.742)	-0.23 (0.413)	-0.16 (0.558)
Best new family	-0.36 (0.191)	-0.58 (0.023)	0.13 (0.648)

RQ-N2, does structure predict the gain? H2 is inconclusive: 0 of 9 correlations are positive at $p < 0.05$ and 1 negative (the hypothesis predicts positive; no multiple-comparison correction; $n = 15$ designs, so only a large ρ can be told from chance). Table VIII lists Spearman's ρ (and p).

Graph size is a confounder. The measure $z(Q)$ rises with the number of edges ($\rho = 0.85$ ($p < 0.001$)), and the gains fall with it: the best new-family gain against edges is $\rho = -0.60$ ($p = 0.017$), the best round-one gain $\rho = -0.29$ ($p = 0.289$), Urd's $\rho = -0.37$ ($p = 0.177$). Table IX repeats the correlations of Table VIII as partial Spearman correlations with the rank of the edge count regressed out of both variables. The null graphs behind $z(Q)$ use 10 double edge swaps per edge on every design (an earlier run capped the swaps at 200000, which left 3.8 to 9.8 per edge on the five graphs above 20000 edges; the cap was removed and the null recomputed). The best-of-k gains are maxima over k families per design ($k = 7$ round one, $k = 4$ round two), a positively biased gain, and the noise of a single VPR seed per configuration is unquantified.

TABLE IX
RQ-N2 with graph size controlled: partial Spearman ρ (p) between structure and seed gain, the rank of the edge count regressed out of both.

Gain	Q	$z(Q)$	$-p_{\mathrm{Rent}}$
Urd, round one	0.11 (0.710)	-0.27 (0.358)	0.18 (0.543)
Best round one	0.15 (0.607)	0.04 (0.891)	-0.14 (0.634)
Best new family	0.09 (0.761)	-0.16 (0.574)	0.24 (0.401)

TABLE X
Round three, headline configurations (the β_m of each source): geomean ratio over the 15 completed designs against the stock packer with the same start, same VPR seeds. Below 1.0 is better; CLBs is the number of logic CLBs; Pack and Flow are the packing and total VPR runtimes. The stock arm ran on the stock vpr and the Norns arms on the Norns vpr, two builds, so Pack and Flow are not a same-binary comparison (the same-binary packing time ratio is in the text). The seed arm's stock reference is the stock packing with the same words_partial seed rebuilt on it.

Arm	WL	Chan.	Delay	CLBs	Pack	Flow
module, analytic	1.051	1.061	1.017	0.965	1.318	1.084
word, analytic	1.032	1.034	1.002	0.985	1.125	1.014
word+io, analytic	1.030	1.036	0.996	0.984	1.130	1.005
module, word seed	1.064	1.063	1.031	0.965	1.315	1.100
word, word seed	1.039	1.042	1.009	0.985	1.116	1.039
word+io, word seed	1.039	1.042	1.007	0.984	1.127	1.052

B. Round three results

RQ-N8, does packing dilute structure? H8 is supported: over the 15 stock packings the median word purity (plurality share) is 0.63 against 0.20 for a random grouping of the same atoms (higher on 15 of 15), and the median split factor is 4.00 (above 1 on 12 of 15). The median module purity is 0.99, so module affinity has little to gain on most designs. Of the logic atoms, 65.8% carry a word (median); 34.2% lack one (front end: ABC's cone provenance, mixed spans 33.8% and no match 0.4%) and a further 20.7% carry a word but sit in a CLB where their word is not the plurality word or fails the 1/3 rule (packing).

RQ-N10, I/O words. H10a is supported: the median LUT word coverage rises from 31.8% to 53.9% with io-words (median gain 24.1%, positive on 15 of 15 designs).

Without the four pilot designs. H8, H10a and H11 use the stock packings and no beta, so the pilot, which only chose beta, cannot affect them; the n=11 pass is reported beside the n=15 status, which stays the primary one. H8 on the 11 designs: supported (median purity 0.59, split 4.00). H10a on the 11 designs: supported (median gain 22.3%, positive on 11 of 11; the rule's 12 of 15 scaled to 9, not registered).

Seed noise. The stock packer's routed wirelength varies across the three VPR seeds with a median coefficient of variation 2.4% (maximum 9.7%, n=15 designs), which is the floor under any wirelength difference below a few percent.

module affinity. H9a is inconclusive (pilot-sensitive): the module-level purity moves by 0.000 (median, n=15; 0.035 without the pilot designs) with a median change of -1.6% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.051 of the stock packer's (better on 3 of 15; 90% interval 1.020 to 1.082; 1.069 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.064 (better on 2 of 15).

word affinity. H9a is supported: the word-level purity moves by 0.111 (median, n=15; 0.120 without the pilot designs) with a median change of 0.0% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.032 of the stock packer's (better on 4 of 15; 90% interval 1.006 to 1.062; 1.053 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.039 (better on 4 of 15).

word+io affinity. H9a is supported: the word+io-level purity moves by 0.104 (median, n=15; 0.124 without the pilot designs) with a median change of 0.0% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.030 of the stock packer's (better on 4 of 15; 90% interval 1.005 to 1.057; 1.050 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.039 (better on 5 of 15).

I/O words in the affinity (H10b). not supported: word+io over word is 0.998 (better on 4 of 15, $p = 1.000$).

Packing time on one binary. The stock arm of Stage 4 ran on the stock vpr and the Norns arms on the Norns vpr, two builds, so the ratios of Table X compare different binaries (and the Flow column includes routing).

Primary same-binary measurement (pack-only style, no routing, seed 1 to 3, 180 runs): on the Norns vpr, for each design and seed the stock arm (option absent) and the three Norns arms ran back to back, so any load affects the arms alike. The geomean over designs of the Norns to stock packing time ratio is 1.305 / 1.120 / 1.129 (module / word / word+io). These runs took place 2026-10-25 10:42:54 to 2026-10-25 11:05:44 (UTC-4) with a concurrent load: another lane ran CPU work outside the heavy lock (the one-minute load average at the start of a run had median 1.2, maximum 4.8). Packing times of the small designs are a few tenths of a second, so the ratio is coarse and approximate.

Not interleaved, hence approximate: the Stage-4 Norns packing times over a separate rerun of the stock arm on the Norns vpr (45 runs, 45 flat placements identical to the Stage-4 stock run, taken at another time and load) give 0.744 / 0.635 / 0.638, against 1.318 / 1.125 / 1.130 with the stock arm on the other build, as first reported.

RQ-N11, what the annealed placement says. H11 is inconclusive under the protocol text (no design excluded; the sign rule needs 12 of the 15 designs, so a relation defined for fewer than 12 designs cannot be supported): same-word pairs: median stratified distance difference -0.176

TABLE XI

Round three: hypotheses, status and deciding number (all designs run; statuses use the rules in docs/plan/round3.md). H9 numbers are for the headline β_m of each source, Holm-adjusted over the three sources; a status that differs between all 15 designs and the 11 without the pilot designs is shown as pilot-sensitive.

RQ	Hypothesis	Status	Deciding number
N8	H8 packing dilutes structure	supported	median word purity 0.63, split 4.00 (n=15)
N10	H10a io-words add coverage	supported	median gain 24.1%, 15/15 designs
N9	H9a module purity up, CLBs within +5%	inconclusive (pilot-sensitive)	purity +0.000 (without pilot +0.035), CLB -1.6%, $p < 0.001$
N9	H9a word purity up, CLBs within +5%	supported	purity +0.111 (without pilot +0.120), CLB 0.0%, $p < 0.001$
N9	H9a word+io purity up, CLBs within +5%	supported	purity +0.104 (without pilot +0.124), CLB 0.0%, $p < 0.001$
N9	H9b module changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.051, 3/15 better, $p = 0.090$; without pilot 1.069, $p = 0.006$
N9	H9b word changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.032, 4/15 better, $p = 0.188$; without pilot 1.053, $p = 0.016$
N9	H9b word+io changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.030, 4/15 better, $p = 0.188$; without pilot 1.050, $p = 0.016$
N9	H9c module+seed beats stock+seed	not supported	geomean 1.064, 2/15 better, $p = 1.000$
N9	H9c word+seed beats stock+seed	not supported	geomean 1.039, 4/15 better, $p = 1.000$
N9	H9c word+io+seed beats stock+seed	not supported	geomean 1.039, 5/15 better, $p = 1.000$
N10	H10b word+io beats word	not supported	geomean 0.998, 4/15 better, $p = 1.000$
N11	H11 same-word pairs closer than nets predict	inconclusive	median -0.176, $p = 0.017$, n=12; 9 of 12 designs sign-consistent, 12 needed; with the unregistered scaling (10 needed): inconclusive
N11	H11 same-module pairs closer than nets predict	inconclusive	median -0.188, $p < 0.001$, n=11; 10 of 11 designs sign-consistent, 12 needed; with the unregistered scaling (9 needed): supported
N11	H11 overall (both relations)	inconclusive	supported only if both are; unregistered scaling: inconclusive

TABLE XII

RQ-N8 per design on the stock packing: logic CLBs, word purity (plurality share), the random-grouping null purity, word split factor (1.0 best), module purity, and the share of logic atoms lost to the front end (no word) and to packing (word atom not in its word's CLB).

Design	CLBs	Purity	Null	Split	Module	Lost: FE	Lost: pack
ch_intrinsics	13	0.86	0.51	3.00	1.00	54.7%	6.1%
stereovision3	15	0.48	0.25	2.00	1.00	40.4%	31.3%
diffeq2	22	0.96	0.31	2.00	1.00	31.8%	7.0%
diffeq1	37	0.80	0.27	7.50	1.00	37.4%	14.2%
mkPktMerge	21	0.86	0.75	1.00	0.92	7.3%	13.2%
raygentop	143	0.52	0.14	3.00	0.68	34.2%	32.6%
sha	142	0.47	0.19	11.50	1.00	50.5%	27.2%
boundtop	54	0.48	0.18	4.00	0.68	33.7%	34.2%
mkSMAdapter4B	125	0.63	0.20	5.00	1.00	51.3%	20.0%
or1200	252	0.64	0.20	5.50	0.76	51.1%	20.7%
stereovision0	707	0.27	0.06	4.00	0.74	22.1%	56.8%
stereovision1	788	0.36	0.08	4.00	0.71	17.1%	53.0%
mkDelayWorker32B	74	0.77	0.47	1.00	0.99	80.7%	4.3%
stereovision2	1956	0.59	0.09	6.00	0.88	4.9%	39.0%
blob_merge	494	0.91	0.09	1.00	1.00	30.5%	8.0%

(one-sided Wilcoxon $p = 0.017$, n=12 designs; negative in at least two of three seeds on 9 designs, 12 needed; status inconclusive); same-module pairs: median stratified distance difference -0.188 (one-sided Wilcoxon $p < 0.001$, n=11 designs; negative in at least two of three seeds on 10 designs, 12 needed; status inconclusive). The variants of Table XIV, labelled: with the sign rule scaled to $\lceil 12n/15 \rceil$ designs, which the first analysis code used and the protocol does not register (the previous primary status): word inconclusive (n=12), module supported (n=11), overall inconclusive; with the 95% exclusion of the amendment d805789: word inconclusive (n=12), module inconclusive (n=11), overall inconclusive (scaled count: word inconclusive (n=12), module supported (n=11), overall inconclusive). Without the four pilot designs the protocol-text statuses are word inconclusive (n=9), module inconclusive (n=9), overall inconclusive; the literal 12 is unreachable there. Neighbour stability across VPR seeds (mean Jaccard of each CLB's four nearest neighbours): 0.267 against 0.058 for a permuted placement. The per-design "why are these neighbours" tables (direct nets, shared high-fanout net, carry chain) are in data/analysis/n11-why/.

TABLE XIII
RQ-N10: I/O words per design (buses of at least two bits) and the LUT word coverage without and with io-words.

Design	io-words	Word cov.	+ io-words
ch_intrinsics	8	16.4%	49.2%
stereovision3	3	26.0%	41.6%
diffeq2	8	31.8%	81.6%
diffeq1	11	21.5%	53.9%
mkPktMerge	5	49.7%	95.8%
raygentop	23	24.4%	57.8%
sha	4	2.5%	47.5%
boundtop	22	42.1%	64.3%
mkSMAdapter4B	29	12.6%	38.0%
or1200	36	27.6%	51.7%
stereovision0	27	32.1%	40.5%
stereovision1	20	50.3%	57.3%
mkDelayWorker32B	27	77.0%	78.1%
stereovision2	24	78.8%	85.2%
blob_merge	4	48.8%	52.2%

C. Round four: results

a) What gets clustered (N17).: Table XVIII gives, per benchmark, the share of CLB member pairs in each class for the stock packing (seed 1; SC series-critical, SN series-noncritical, SI shared-input, OR orthogonal, FL filler), the SC share of the headline Norns packing, the chance level of SC+SN under 50 null packings, and the near-critical paths within 10% of the critical path (k) and how many of them are split across CLBs (m). Per benchmark:

- ch_intrinsics: 1.6% shared-input, 34.8% orthogonal, 2.5% series-critical, 61.0% SN+FL (non-critical remainder: SN 4.9%, FL 56.1%); 5 near-critical reported paths (within 10%), 1 of them split across CLBs.
- stereovision3: 12.1% shared-input, 11.4% orthogonal, 5.6% series-critical, 70.8% SN+FL (non-critical remainder: SN 2.7%, FL 68.1%); 1 near-critical reported path (within 10%), 1 of them split across CLBs.
- diffeq2: 1.1% shared-input, 63.4% orthogonal, 4.3% series-critical, 31.1% SN+FL (non-critical remainder: SN 3.2%, FL 27.9%); 32 near-critical reported paths (within 10%), 32 of them split across CLBs.
- diffeq1: 1.6% shared-input, 53.3% orthogonal, 2.8% series-critical, 42.3% SN+FL (non-critical remainder: SN 4.9%, FL 37.4%); 32 near-critical reported paths (within 10%), 32 of them split across CLBs.
- mkPktMerge: 5.2% shared-input, 59.4% orthogonal, 2.9% series-critical, 32.5% SN+FL (non-critical remainder: SN 1.5%, FL 31.1%); 39 near-critical reported paths (within 10%), 0 of them split across CLBs.
- raygentop: 4.5% shared-input, 25.3% orthogonal, 5.0% series-critical, 65.2% SN+FL (non-critical remainder: SN 3.1%, FL 62.0%); 55 near-critical reported paths (within 10%), 9 of them split across CLBs.
- sha: 10.7% shared-input, 21.3% orthogonal, 1.4% series-critical, 66.6% SN+FL (non-critical remainder: SN 9.1%, FL 57.6%); 26 near-critical reported paths (within 10%), 26 of them split across CLBs.
- boundtop: 3.5% shared-input, 26.6% orthogonal, 5.0% series-critical, 64.9% SN+FL (non-critical remainder: SN 2.5%, FL 62.4%); 122 near-critical reported paths (within 10%), 118 of them split across CLBs.
- mkSMAdapter4B: 5.8% shared-input, 24.9% orthogonal, 4.9% series-critical, 64.4% SN+FL (non-critical remainder: SN 5.7%, FL 58.7%); 7 near-critical reported paths (within 10%), 7 of them split across CLBs.
- or1200: 7.9% shared-input, 30.5% orthogonal, 7.8% series-critical, 53.8% SN+FL (non-critical remainder: SN 2.4%, FL 51.4%); 72 near-critical reported paths (within 10%), 72 of them split across CLBs.
- stereovision0: 2.1% shared-input, 16.2% orthogonal, 3.1% series-critical, 78.6% SN+FL (non-critical remainder: SN 2.8%, FL 75.8%); 1 near-critical reported path (within 10%), 1 of them split across CLBs.
- stereovision1: 1.7% shared-input, 26.5% orthogonal, 4.0% series-critical, 67.8% SN+FL (non-critical remainder: SN 1.4%, FL 66.4%); 51 near-critical reported paths (within 10%), 0 of them split across CLBs.
- mkDelayWorker32B: 5.4% shared-input, 17.8% orthogonal, 8.9% series-critical, 67.9% SN+FL (non-critical remainder: SN 3.9%, FL 63.9%); 1 near-critical reported path (within 10%), 1 of them split across CLBs.
- stereovision2: 1.3% shared-input, 40.2% orthogonal, 3.4% series-critical, 55.1% SN+FL (non-critical remainder: SN 3.7%, FL 51.4%); 431 near-critical reported paths (within 10%), 431 of them split across CLBs.
- blob_merge: 1.7% shared-input, 51.6% orthogonal, 3.3% series-critical, 43.3% SN+FL (non-critical remainder: SN 5.9%, FL 37.5%); 3 near-critical reported paths (within 10%), 3 of them split across CLBs.

Table XVI weights the pairs by criticality (median over designs) and Table XVII gives the near-critical counts at both margins and the reasons for the crossings; per-design values are in data/analysis/r4-results.json and data/<design>/r4near.json. Source shares of the molecules added: stock connectivity_timing 67.0%, high_fanout 1.4%, seed 9.8%, transitive 21.8%.

b) Knowns (N12).: Table XX: 2 relations meet the literal rule (at least 12 of 15 designs, and the stated test); 3 with the count scaled to the designs defined (labelled, not registered). K1 is defined on 11 designs, so the literal count of 12 is unreachable by construction and its "no" is guaranteed whatever the data; the scaled column is a labelled variant. K3 is two registered conditions: A (share ≤ 0.80) holds in 11/15 designs (needs 12) and B (above the null) in 15/15 (needs all 15); it is known iff both hold. K8 is registered across the seeds within a design: per packing it holds for 10/15 designs (stock) and 10/15 (norns), known iff both reach 12; the pooled row (6 points per design, both packings) is a labelled variant and is not counted.

c) Hierarchy and chip size (N13).: The designs are small (n=15). Provenance names carry one module/leaf@file:line level, so the depth D read from them is 1 in every design by construction: depth is not measurable from the provenance names, and H13 as registered tests the module count M only. The per-level Rent exponent is therefore a single-level fit, defined for 7 designs and empty for 8; it is not reported further. Exploratory (not registered; instances and depth of Odin III's elaborated design before flattening against log grid area): norns packing, depth: $\rho = 0.67$, Holm $p = 0.023$; norns packing, instances: $\rho = 0.74$, Holm $p = 0.009$; norns packing,

TABLE XIV

RQ-N11, H11 under the protocol text and the labelled variants. n is the number of designs in which the relation is defined (and not excluded); median is the median over those designs of the stratified distance difference; k/need is the number of designs negative in two of three seeds over the number the rule needs; perm p is the permutation p of the median over designs (1000 label permutations pooled over designs for the protocol text and the scaled count, 200 for the amendment), reported and not used in the status. The status uses the median, the one-sided Wilcoxon test and k/need. The count of the pilot-free rows is shown scaled in the scaled rows only; with 11 designs the literal 12 cannot be met.

	Same word					Same module				
Variant	n	median	k/need	perm p	status	n	median	k/need	perm p	status
All 15 designs										
Protocol text, 12 of 15 designs (primary)	12	-0.176	9/12	<0.001	inconclusive	11	-0.188	10/12	<0.001	inconclusive
Scaled count (not registered)	12	-0.176	9/10	<0.001	inconclusive	11	-0.188	10/9	<0.001	supported
Amendment d805789, 12 of 15	12	-0.176	9/12	0.005	inconclusive	11	-0.188	10/12	0.005	inconclusive
Amendment d805789, scaled count	12	-0.176	9/10	0.005	inconclusive	11	-0.188	10/9	0.005	supported
Without the four pilot designs										
Protocol text, 12 of 15 designs (primary)	9	-0.127	6/12	<0.001	inconclusive	9	-0.141	8/12	<0.001	inconclusive
Scaled count (not registered)	9	-0.127	6/8	<0.001	inconclusive	9	-0.141	8/8	<0.001	supported
Amendment d805789, 12 of 15	9	-0.127	6/12	0.005	inconclusive	9	-0.141	8/12	0.005	inconclusive
Amendment d805789, scaled count	9	-0.127	6/8	0.005	inconclusive	9	-0.141	8/8	0.005	supported

TABLE XV

Packer diagnostics per design at the headline β (seed 1). Matched: share of affinity-file entries that named an atom of the netlist (from the VPR log; the lowest per source: module 29.9% on mkDelayWorker32B; word 2.3% on mkDelayWorker32B; word+io 2.3% on mkDelayWorker32B). Queued: share of molecules handed to the candidate queue by the proposal step that entered it. 0-queued: share of proposal calls (one per cluster) that queued nothing. Columns in each group: module, word, word+io.

	Matched			Queued			0-queued		
Design	Mod.	Word	W+io	Mod.	Word	W+io	Mod.	Word	W+io
ch_intrinsics	100.0%	71.7%	63.8%	81.7%	86.9%	86.9%	1.0%	19.8%	19.8%
stereovision3	100.0%	100.0%	98.9%	66.3%	11.8%	11.8%	5.1%	84.2%	84.2%
diffeq2	86.1%	100.0%	100.0%	63.4%	74.7%	74.7%	6.1%	19.6%	22.0%
diffeq1	92.4%	100.0%	100.0%	53.2%	59.1%	59.1%	7.1%	27.8%	27.8%
mkPktMerge	100.0%	100.0%	54.1%	21.9%	23.8%	23.8%	58.2%	57.5%	57.5%
raygentop	95.1%	96.1%	96.4%	48.8%	42.9%	38.1%	13.1%	59.5%	59.8%
sha	94.4%	100.0%	100.0%	88.0%	45.6%	47.8%	4.1%	61.6%	57.0%
boundtop	90.2%	89.5%	90.1%	44.2%	63.0%	64.1%	15.9%	49.0%	47.6%
mkSMAdapter4B	96.5%	85.5%	88.7%	21.9%	49.6%	47.6%	20.5%	54.6%	47.4%
or1200	97.7%	97.6%	96.0%	29.1%	46.4%	42.1%	22.0%	52.4%	49.2%
stereovision0	100.0%	100.0%	99.8%	79.7%	77.6%	77.8%	6.8%	68.0%	67.8%
stereovision1	99.4%	99.3%	99.2%	34.1%	77.8%	76.0%	41.3%	44.4%	43.8%
mkDelayWorker32B	29.9%	2.3%	2.3%	7.9%	0.0%	0.0%	60.4%	100.0%	100.0%
stereovision2	73.0%	100.0%	99.9%	17.5%	46.1%	46.2%	57.4%	37.2%	37.0%
blob_merge	99.8%	100.0%	99.0%	92.4%	5.4%	12.9%	0.3%	98.5%	97.1%

TABLE XVI

Median class shares (percent) over designs, unweighted and weighted by pack-time (w_1, minimum atom criticality) and post-route (w_2, minimum path-criticality ratio) criticality; n = designs (w2 needs a routed run whose packing equals the census packing).

packing	weight	SC	SN	SI	OR	FL	n
stock	unweighted	4.0	3.2	3.5	26.6	57.6	15
stock	pack-time w_1	5.0	2.6	3.6	28.6	56.4	15
stock	post-route w_2	5.5	2.7	4.1	29.8	54.1	15
norns	unweighted	3.8	2.9	3.7	38.7	51.3	15
norns	pack-time w_1	4.8	2.1	3.6	38.5	49.1	15
norns	post-route w_2	4.9	1.7	6.7	38.5	45.2	15

TABLE XVII

Near-critical reported paths at the 5% and 10% margins (seed 1, routed run): designs, median count k, designs with at least one path split across CLBs, member-member hop crossings and hops (pooled), and the reason of each crossing (R1 refused by the legalizer, R2 queued and not taken, R3 never proposed, R0 no census record).

packing	margin	designs	median k	with split	crossings/hops	R1	R2	R3	R0
stock	5%	15	10	13	1761/7102	317	47	1397	0
stock	10%	15	32	13	3608/13548	606	78	2924	0
norns	5%	15	14	13	1003/4176	226	68	709	0
norns	10%	15	19	13	4260/13262	719	111	3430	0

instances given design size: $\rho = 0.58$, Holm $p = 0.056$; stock packing, depth: $\rho = 0.67$, Holm $p = 0.023$; stock packing, instances: $\rho = 0.74$, Holm $p = 0.009$; stock packing, instances given design size: $\rho = 0.58$, Holm $p = 0.056$ (Holm over 6 tests; depth runs from 1 to 5 modules). Table XXI lists atoms S, modules M, depth D from the names, the instance count and depth of the elaborated design (exploratory), and the mean grid area under stock and norns packing.

TABLE XVIII
Pair classes in the stock packing (percent of pairs), headline-packer SC share, chance SC+SN, near-critical paths (k) and split ones (m).

design	CLBs	SC	SN	SI	OR	FL	SC(r4)	null SC+SN	k	m
ch_intrinsics	13	2.5	4.9	1.6	34.8	56.1	2.1	1.0	5	1
stereovision3	15	5.6	2.7	12.1	11.4	68.1	5.1	1.3	1	1
diffeq2	22	4.3	3.2	1.1	63.4	27.9	4.3	0.7	32	32
diffeq1	37	2.8	4.9	1.6	53.3	37.4	2.7	0.4	32	32
mkPktMerge	21	2.9	1.5	5.2	59.4	31.1	2.9	0.3	39	0
raygentop	143	5.0	3.1	4.5	25.3	62.0	4.3	0.1	55	9
sha	142	1.4	9.1	10.7	21.3	57.6	1.0	0.1	26	26
boundtop	54	5.0	2.5	3.5	26.6	62.4	4.2	0.3	122	108
mkSMAdapter4B	125	4.9	5.7	5.8	24.9	58.7	4.3	0.2	7	7
or1200	252	7.8	2.4	7.9	30.5	51.4	6.6	0.1	72	72
stereovision0	707	3.1	2.8	2.1	16.2	75.8	3.0	0.0	1	1
stereovision1	788	4.0	1.4	1.7	26.5	66.4	3.8	0.0	51	0
mkDelayWorker32B	74	8.9	3.9	5.4	17.8	63.9	8.9	0.4	1	1
stereovision2	1956	3.4	3.7	1.3	40.2	51.4	3.3	0.0	431	431
blob_merge	494	3.3	5.9	1.7	51.6	37.5	3.3	0.0	3	3

d) Packing and placement start with critical-path delay (N14, N16).: Table XXII gives the geometric mean of per-design ratios (3 seeds); below 1 is better. Checks: VPR's log CPD equals path 1 of the timing report (arrival minus required time, i.e. minus its slack) within 0.0005 ns (report precision) in 264 of 264 runs (the data arrival time alone is 0.023 ns below the CPD by median, at most 0.467 ns: the setup time); the attribution's own CPD is read from the same log line and is not an independent check; the stock arm reproduces round three's WL, W and CPD in 45 of 45 runs.

D. Round five: results

Table XXIII gives every hypothesis with its status and deciding number. Table XXIV gives the geometric-mean ratio of each packer to the known clustering per size (routed wirelength, then logic CLB count). The netlist level is the headroom number; the HDL level is the Norns question. Variants (labelled, no status changed): sizes $B \geq 32$ only: NL stock CLB 1.072, WL 1.237, CPD 1.109 (supported for WL); HDL stock CLB 1.073, WL 1.237, CPD 1.089 (supported for WL); HDL norns CLB 1.027, WL 1.091, CPD 1.022 (inconclusive for WL). Reported, not tested: the channel-width ratio and the cut ratio of each packer (NL stock: W 1.026, cut 1.126; HDL stock: W 1.030, cut 1.127; HDL norns: W 1.014, cut 1.061); the largest null purity of any packing against the known blocks was 0.438. The Odin III attribution agreed with VPR's critical-path delay (0.5% tolerance) in 999 of 999 runs; the share of member-to-member hops on the reported critical paths that cross CLBs is HDL_known 42.0%, HDL_norns 43.0%, HDL_stock 43.6%, NL_known 43.1%, NL_stock 44.3%.

a) Exploratory (dated entry (e), no status changed): by topology.: Ratio to the known clustering by topology (geometric mean over designs): NL_stock_WL: chain 1.004, mesh 1.423, rand 1.019, tree 1.522; HDL_stock_WL: chain 1.027, mesh 1.444, rand 1.011, tree 1.528; HDL_norns_WL: chain 1.028, mesh 1.168, rand 1.009, tree 1.224; NL_stock_CLB: chain 1.001, mesh 1.169, rand 1.007, tree 1.124; HDL_stock_CLB: chain 1.003, mesh 1.178, rand 1.007, tree 1.128; HDL_norns_CLB: chain 1.003, mesh 1.063, rand 1.007, tree 1.059.

b) Exploratory, computed (dated entry (f), no status changed): promised variants.: Fraction of the gap closed by the HDL-aware packer, $(r_{stock} - r_{norns}) / (r_{stock} - 1)$, on HDL designs with $r_{stock} \geq 1.03$: CLB mean 0.540, median 1.000, n=29; CPD mean 0.423, median 0.131, n=34; WL mean 0.534, median 0.878, n=32. The H19/H20/H25 rule applied per topology (wirelength): NL stock: chain not supported (WL 1.004, n=18), mesh supported (WL 1.423, n=18), rand not supported (WL 1.019, n=18), tree supported (WL 1.522, n=18); HDL stock: chain inconclusive (WL 1.027, n=15), mesh supported (WL 1.444, n=15), rand not supported (WL 1.011, n=18), tree supported (WL 1.528, n=15); HDL norns: chain inconclusive (WL 1.028, n=15), mesh supported (WL 1.168, n=15), rand not supported (WL 1.009, n=18), tree supported (WL 1.224, n=15). H25 on the 37 netlist designs that pass the literal zero-violation check (H28): CLB 1.059 (supported), WL 1.188 (supported), CPD 1.054 (supported); those designs are mostly chains and trees (mesh and random graphs rarely pass), so this is not the same population. Leak measurement: in dp the Norns .net equals the stock .net on all three seeds for 6 of 36 designs (mean purity against the blocks 0.9998 against 0.9501); in ctl for 27 of 27 (purity 0.9384 against 0.9384).

V. Discussion

Three observations. First, the seed families span a narrow band but are not interchangeable: provenance (Urd) is the worst full seed in the soft arm (1.029) and the best in the hard arm, so it is not consistently better or worse than recovered clusters. The full and inter-cluster-only arms differ by 0.6% in pooled wirelength (1.009 vs 1.015); round one has no arm that randomizes the coarse cluster layout; the round-two permuted-cluster and random-start arms (Section IV-A) measure it. Second, VPR's checkpointing does not separate the crossover in the measured range: the two sweeps find the same scale on 14 of the 15 designs where both found one, and the control extends it on stereovision0 (control 32 vs 16). RLPlace's prediction that checkpointing extends the useful range is therefore not observed on these designs; if it holds, its effect is smaller than the between-design variation. Third, the safe recipe is to cold-start from a cluster-derived placement at a temperature scale at or below the crossover, which is a multiple of VPR's own estimate; above it the seed's retention is still high per family (medians 0.52-0.74 at twice the crossover, individual runs 0.21-1.00) but it wins only 57 of 105 head-to-heads against an unseeded run at the same temperature – the coarse structure survives the anneal, the advantage does not.

TABLE XIX
Round-four hypotheses (n=15 unless stated; literal rules of docs/plan/round4.md).

RQ	hypothesis	status	deciding number
N17	H17a shape (10/75/15)	not supported	medians SC 4.0%, SI+OR 32.0%, SN+FL 64.4%, n=15
N17	H17a literal numbers	not supported	same medians against 10/75/15 ±0.10
N17	H17b packer changes SC share	not supported	median ΔSC -0.003, Holm $p < 0.001$, n=15
N17	H17c critical edges kept together	supported	median ratio 0.652, Wilcoxon $p = 0.027$, n=9 (literal rule; reading with an n<12 clause: inconclusive (n < 12))
N17	H17d series share above the atom-shuffle null (not a clustering-quality result)	supported	median excess 0.072, 15/15 positive; exploratory molecule-unit null: excess 0.023, p<0.001, inconclusive
N13	H13a module count vs grid area	supported	n=15 (M only: depth not measurable from provenance names)
N13	H13b beyond design size	inconclusive	n=15
N13	H13c packer effect vs hierarchy	inconclusive	n=15
N15	H15 depth estimate predicts critical statements	inconclusive	median ρ 0.137, n=12
N14	H14 norns vs stock: wirelength	not supported	geomean 1.033, 4/15 better, Holm $p = 1.000$
N14	H14 norns vs stock: chan_width	not supported	geomean 1.015, 4/15 better, Holm $p = 1.000$
N14	H14 norns vs stock: cpd_ns	not supported	geomean 0.996, 6/15 better, Holm $p = 1.000$
N16	H16 stage×bit seed: norns.chan_width	inconclusive	geomean 0.974, 8/14 better, Holm $p = 0.684$
N16	H16 stage×bit seed: norns.cpd_ns	not supported	geomean 1.013, 6/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: norns.wirelength	not supported	geomean 1.015, 6/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: stock.chan_width	inconclusive	geomean 0.977, 10/14 better, Holm $p = 0.581$
N16	H16 stage×bit seed: stock.cpd_ns	inconclusive	geomean 0.996, 7/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: stock.wirelength	not supported	geomean 1.005, 6/14 better, Holm $p = 1.000$

TABLE XX
Knowns from the dataset: holds in k of n designs, test p, known under the literal rule and under the scaled count.

	relation	holds	p	known	scaled
K1	same-module CLBs closer than net connectivity predicts (all 3 seeds)	10/11	<0.001	no	yes
K2	same-word CLBs closer than net connectivity predicts (all 3 seeds)	9/12	0.017	no	no
K3	stock packing dilutes words: word plurality share ≤ 0.80 (A, in ≥ 12 designs) and above the null's (B, in all 15)	11/15	<0.001	no	no
K4	4-nearest-neighbour stability across seeds exceeds the permuted null	15/15	<0.001	yes	yes
K5	packing does not depend on the VPR seed (.net identical across the 3 seeds)	30/30	n/a	yes	yes
K6	the grid is sized by the logic (clb) tile: it has the highest utilisation	4/15	n/a	no	no
K7	adder, multiplier and memory atoms hold more of the critical-path delay than of the design's atoms	11/15	<0.001	no	no
K8 stock	routed wirelength and CPD move together across the seeds of one design (Spearman > 0) [stock packing]	10/15	n/a	no	no
K8 norns	routed wirelength and CPD move together across the seeds of one design (Spearman > 0) [norns packing]	10/15	n/a	no	no
K8 pooled	routed wirelength and CPD move together across the seeds of one design (Spearman > 0) [labelled variant: both packings pooled, 6 points]	11/15	n/a	no	no

Round two. The bitwise idea is tested in four forms, and the evidence is this: word clusters against module clusters are inconclusive (0.986); laying the words out bit-sliced against the quotient layout of the same clusters is inconclusive (0.982 in wirelength); the hierarchical combination is not supported as the best seed (rank 8 of 11); and seeding only the word cells is not distinguishable from seeding everything (H7d inconclusive, 0.980, better on 9 of 15). The word membership available from Odin III covers the flip-flops, adders, multipliers and memories but only 49% of the LUTs, because ABC's cone provenance merges the spans of mixed logic, so the word seeds are datapath-register seeds more than whole-datapath seeds; a result against bitwise seeding here is a result about that coverage as much as about the idea.

Every soft arm of Table VI, bitwise or not, random start included, lands between 0.980 and 1.029 of the baseline's wirelength: a band of 4.9% in a geomean over 15 designs. The conclusions below are conclusions about differences inside that band.

One confound the scale-1 comparison cannot remove: VPR derives its start temperature T_0 from the starting placement, so a start that is worse than the analytic one is annealed hotter and, in these runs, longer. The geomean start temperature relative to the baseline is 9.24 for the random start, 3.23 for Urd and 4.63 for the best new family, and placement runtime moves with it (Table VI). The arms are therefore compared at VPR's own temperature, not at equal anneal effort. The confound bears directly on four comparisons: H4 (random start, permuted and quotient layouts have T_0 of 9.24, 3.09 and 3.23 times the baseline's, for Urd), H7a (words 6.24 against Urd 3.23), H7b (slices 8.07 against words 6.24) and H7d (words_partial 4.63 against words 6.24; against the permuted Urd arm 3.09). The start temperatures of the arms differ by 3.1 to 9.2 times the baseline's, and the

TABLE XXI
Hierarchy and grid area (mean of 3 seeds, analytic start). M, D: from provenance names; instances, depth: elaborated design before flattening (exploratory).

design	S	M	D	instances	depth	area stock	area norns
ch_intrinsics	180	1	1	2	2	100.00	100.00
stereovision3	319	1	1	1	1	49.00	49.00
diffeq2	429	1	1	1	1	256.00	256.00
diffeq1	622	1	1	1	1	256.00	256.00
mkPktMerge	220	4	1	7	3	784.00	784.00
raygentop	2439	14	1	16	3	361.00	361.00
sha	2495	1	1	1	1	289.00	289.00
boundtop	930	8	1	16	3	225.00	225.00
mkSMAdapter4B	2010	2	1	11	3	400.00	400.00
or1200	3585	13	1	16	3	729.00	729.00
stereovision0	21002	23	1	302	4	1089.00	1156.00
stereovision1	18845	13	1	233	5	1600.00	1600.00
mkDelayWorker32B	541	2	1	19	3	2500.00	2500.00
stereovision2	36057	20	1	146	3	6724.00	6724.00
blob_merge	10075	2	1	3	2	784.00	784.00

TABLE XXII
Arm ratios (geomean over designs).

metric	comparison	geomean	better
wirelength	norns/stock, analytic	1.033	4/15
wirelength	norns: stagebit/analytic	1.015	6/14
wirelength	stock: stagebit/analytic	1.005	6/14
wirelength	norns.stagebit_vs_wordseed	0.995	4/14
wirelength	norns.wordseed_vs_analytic	1.018	7/15
wirelength	stock.stagebit_vs_wordseed	0.998	9/14
wirelength	stock.wordseed_vs_analytic	1.007	7/15
chan_width	norns/stock, analytic	1.015	4/15
chan_width	norns: stagebit/analytic	0.974	8/14
chan_width	stock: stagebit/analytic	0.977	10/14
chan_width	norns.stagebit_vs_wordseed	0.998	6/14
chan_width	norns.wordseed_vs_analytic	0.977	10/15
chan_width	stock.stagebit_vs_wordseed	1.018	4/14
chan_width	stock.wordseed_vs_analytic	0.963	9/15
cpd_ns	norns/stock, analytic	0.996	6/15
cpd_ns	norns: stagebit/analytic	1.013	6/14
cpd_ns	stock: stagebit/analytic	0.996	7/14
cpd_ns	norns.stagebit_vs_wordseed	1.010	7/14
cpd_ns	norns.wordseed_vs_analytic	1.002	8/15
cpd_ns	stock.stagebit_vs_wordseed	1.004	7/14
cpd_ns	stock.wordseed_vs_analytic	0.992	10/15

direction of the bias is unknown: a hotter start can erase more of the seed, or give the anneal more moves to repair a worse one. A random start beating a connectivity-aware start would be a statement about the schedule as much as about the start. The effort curve (N3) is the experiment that separates the two, and it stays open.

On the coarse-versus-fine question that round one left open, the arms give these numbers: H4 is not supported, with Urd at random 0.997, permuted 1.020 and quotient 1.029. RQ-N1 measures part of the round-one parity: VPR's final placement has 2.78× the null's module compactness and 3.79× its word compactness (medians), which is inconclusive as a rediscovery of the hierarchy by the NMI-and-compactness rule, which suggests the placer already groups much of what a module or word seed would group. Whether the structure of the netlist predicts where a seed helps is H2, inconclusive (the one correlation that reaches $p < 0.05$ is negative: the best new-family gain vs $z(Q)$; it tracks graph size, and with the edge count regressed out the partial ρ is -0.16, $p = 0.574$). All of this is $n = 15$ designs, median 125 CLBs, one VPR seed per configuration; the seed-to-seed spread of VPR is not measured here, so differences of about a percent should not be read as effects.

A. Round three discussion

Packing does dilute word structure (H8 supported): a word's atoms are spread over about 4.00 times the CLBs they would need, and the purity of the stock CLBs (0.63) is well above a random grouping (0.20) but far from word-pure. Of the logic atoms, 34.2% have no word at all (the front end) and a further 20.7% are word atoms packed away from their word (packing); these are two separate per-design medians, and the median front-end loss is larger than the median packing loss.

I/O words are a cheap way to recover part of the front-end loss (H10a supported: LUT word coverage 31.8% to 53.9%).

The Norns packer does what it was built to do: with the word affinity the word purity rises by 0.111 at a median CLB change of 0.0% (H9a supported; module inconclusive (pilot-sensitive), word+io supported). The module source has almost nothing to act on, because the designs are close to one module each (median module purity 0.99).

Purer clusters did not give shorter wires. With the analytic start the word-affinity packing routes at 1.032 of the stock packer's wirelength (better on 4 of 15 designs, 90% interval 1.006 to 1.062), so H9b is inconclusive (pilot-sensitive): the sign is positive (worse) on every source, the two-sided test reaches the 5% level only without the four pilot designs, and the status is therefore not stated as a change. With the round-two word seed rebuilt on the new packing the ratio is 1.039 (H9c not supported): giving the round-two seed word-pure CLBs does not rescue it. The packing time grows by a factor 1.125 (word) and 1.318 (module). The seed noise (median coefficient of variation 2.4%, up to 9.7%) is of the same order as the effect, which is why the statistics use the design as the unit with three seeds each.

These runs do not remove the start-temperature confound of round two for the seeded arms, because a seeded start still carries its own start temperature; the analytic-start arm (H9b) does not have it, since the packing is the only difference.

In the median design both relations sit closer than their direct nets alone predict (stratified distance difference -0.176 for same-word pairs, -0.188 for same-module pairs), but not on every design: the same-word difference is positive on 3 designs (boundtop, raygentop, stereovision2) and the same-module difference on 1 designs (blob_merge). H11 is inconclusive under the protocol text, because its rule needs both relations to pass: same word, n=12, status inconclusive (negative in two of three seeds on 9 designs, 12 needed); same module, n=11, status inconclusive (10 designs, 12 needed). Two deviations of the analysis from

TABLE XXIII
Round-five hypotheses (protocol 7886f49). A star marks a labelled variant whose rule was introduced after data; H25 to H27 are decided on the designs passing the starred H28 rule. Ratios are packer / known clustering, geometric mean over designs of the geometric mean over three seeds.

H	test	status	deciding number
H18	HDL: synthesis keeps the construction (literal: original label rule)	not supported	32 of 72 kept (44.4%)
H18*	variant: amended label rule (entry (a), after the labels were seen)	inconclusive	63 of 72 kept (87.5%)
H28	NL: netlist check passes (literal: entry (b), zero improving swaps)	not supported	37 of 72 pass (51.4%)
H28*	variant: slack $\leq$ 10% (entry (c), chosen after the data)	supported	72 of 72 pass (100.0%)
H28*	variant: slack $\leq$ 5% (sensitivity)	inconclusive	63 of 72 pass (87.5%)
H24	HDL: forced clustering reproduces the blocks	supported	mean reproduction 1.000, 100.0% of runs exact, n=189 runs
H27	NL: forced clustering reproduces the blocks	supported	mean reproduction 1.000, 100.0% of runs exact, n=216 runs
H25	NL stock vs known: CLB	supported	geomean 1.073, 90% CI [1.056, 1.090], Holm $p < 0.001$, n=72
H25	NL stock vs known: WL	supported	geomean 1.220, 90% CI [1.168, 1.275], Holm $p < 0.001$, n=72
H25	NL stock vs known: CPD	supported	geomean 1.093, 90% CI [1.068, 1.118], Holm $p < 0.001$, n=72
H19	HDL stock vs known: CLB	supported	geomean 1.073, 90% CI [1.055, 1.092], Holm $p < 0.001$, n=63
H19	HDL stock vs known: WL	supported	geomean 1.219, 90% CI [1.163, 1.280], Holm $p < 0.001$, n=63
H19	HDL stock vs known: CPD	supported	geomean 1.082, 90% CI [1.055, 1.110], Holm $p < 0.001$, n=63
H20	HDL norns vs known: CLB	inconclusive	geomean 1.031, 90% CI [1.019, 1.045], Holm $p < 0.001$, n=63
H20	HDL norns vs known: WL	supported	geomean 1.099, 90% CI [1.057, 1.147], Holm $p = 0.003$, n=63
H20	HDL norns vs known: CPD	inconclusive	geomean 1.032, 90% CI [1.009, 1.056], Holm $p = 0.018$, n=63
H26	NL stock: CLB	not supported	median slope 0.000 per doubling, 5/8 cells positive, Holm $p = 0.578$, permutation $p = 0.236$
H26	NL stock: WL	inconclusive	median slope 0.004 per doubling, 5/8 cells positive, Holm $p = 0.461$, permutation $p = 0.070$
H21	HDL stock: CLB	not supported	median slope 0.000 per doubling, 5/8 cells positive, Holm $p = 1.000$, permutation $p = 0.335$
H21	HDL stock: WL	inconclusive	median slope 0.009 per doubling, 6/8 cells positive, Holm $p = 0.391$, permutation $p = 0.051$
H21	HDL norns: CLB	not supported	median slope 0.000 per doubling, 2/8 cells positive, Holm $p = 1.000$, permutation $p = 0.797$
H21	HDL norns: WL	not supported	median slope -0.001 per doubling, 3/8 cells positive, Holm $p = 1.000$, permutation $p = 0.716$
H22	HDL norns vs stock: CLB	supported	geomean 0.961, 90% CI [0.946, 0.976], Holm $p < 0.001$, n=63
H22	HDL norns vs stock: WL	supported	geomean 0.902, 90% CI [0.866, 0.936], Holm $p < 0.001$, n=63
H22	HDL norns vs stock: CPD	supported	geomean 0.954, 90% CI [0.935, 0.971], Holm $p < 0.001$, n=63
H23	purity gain, dp minus ctl	not supported	median 0.000, $p < 0.001$, 51.9% of pairs positive, n=27 pairs

TABLE XXIV
Gap to the known clustering per size B: routed wirelength ratio (left three) and logic CLB ratio (right three); NL stock, HDL stock, HDL norns.

B	NL	stock	norns	NL	stock	norns
4	1.073	1.052	1.029	1.057	1.057	1.028
8	1.237	1.255	1.151	1.102	1.102	1.059
16	1.262	1.275	1.158	1.061	1.061	1.030
32	1.223	1.220	1.111	1.081	1.080	1.034
64	1.301	1.353	1.153	1.079	1.099	1.046
128	1.286	1.314	1.156	1.074	1.083	1.042
256	1.224	1.204	1.089	1.074	1.066	1.021
512	1.210	1.152	0.999	1.071	1.049	1.000
1024	1.182	1.125	0.987	1.055	1.045	1.000

the protocol text are reported beside it, not instead of it: the unregistered scaling of the count to $\lceil 12n/15 \rceil$ designs, which the first analysis used (r3stats.py), gives word inconclusive (n=12), module supported (n=11), overall inconclusive; the amendment d805789, which excludes a design from a relation when 95% or more of its sampled pairs are related, for the word relation as well as the module relation, gives word inconclusive (n=12), module inconclusive (n=11), overall inconclusive. Both are recorded in the entry of 2026-10-25 of docs/plan/round3.md and in the Method. The neighbours are only partly stable: the mean Jaccard of each CLB's four nearest neighbours between VPR seeds is 0.267, against 0.058 for a permuted placement.

Round three leaves open the effort curve (N3), larger designs (N5), region looseness (N6), a seed-selector change in the Norns packer, and whether a different gain scale (the grid ended at the largest beta of the pilot list; the pilot did not discriminate the betas, so the grid is not informed by it) changes the sign of H9b. The packing time ratios of Stage 4 compare two builds and the proposal step has the counting and ordering limits described in the Method, both to be fixed in round 4.

B. Round four: discussion

The census answers Peter's question with data. In the stock packing the pairs that are neither series-critical nor shared-input nor orthogonal are 64.4% by median (SN+FL, the non-critical remainder: series-noncritical 3.2%, filler 57.6%; a series-noncritical pair does share a direct net, so this is not a share of pairs without a net). Series-critical pairs are 4.0% and shared-input plus

orthogonal pairs 32.0%; the illustration of 10/75/15 is not supported in shape and not supported in numbers. The orthogonal class counts same-word pairs, and words cover about half the LUTs, so it is a lower bound; "not supported" for the shape rests partly on that bound (the rule's $t < 0.40$ arm). The headline packer changes the series-critical share by -0.003 (median). Near-critical paths are counted per benchmark (reported paths) in Table XVIII; most designs have few reported paths within 10%, so H17c rests on 9 designs that pass the registered 20-hop filter: by the literal rule it is supported (median ratio 0.652, one-sided Wilcoxon $p = 0.027$); under the labelled reading with an n<12 clause (not in the registered rule) it is inconclusive (n < 12). H17d is supported against the registered null, which shuffles atoms (excess 0.072 of series pairs over the null's 0.2%, observed 7.7%). That null ignores that the prepacker forces a LUT and its FF into one CLB and that carry chains fill CLBs in sequence. An exploratory null that keeps those molecules as units and leaves the 21.6% (median) of CLBs that hold an adder as they are gives a null of 5.3% and an excess of 0.023 (inconclusive by the same rule, p<0.001). H17d therefore says that the packed pairs include forced ones; it is not a measure of clustering quality. The hierarchy test (N13) is on 15 small designs, and the provenance names expose one level, so it tests the module count, not depth. The round-three affinity files lacked every adder atom; the round-four headline packing is therefore not the round-three packing, and round three's purity and wirelength results describe a packer that did not see carry chains. Limits: the depth estimate is a text heuristic; the attribution workaround uses the tool's functions on BLIFs that carry provenance names; the packing is APPack inside the analytical flow.

C. Round five: discussion

At the netlist level the stock packer's wirelength is 1.220 times, its CLB count 1.073 times and its critical-path delay 1.093 times those obtained from the known clustering (statuses: WL supported, CLB supported, CPD supported), on the designs passing the slack rule of entry (c); the literal exactness check (H28) is not supported, so the cut is not shown to be minimal and only the CLB count is exact. The wirelength gap changes by 0.004 (natural-log ratio) per doubling of the block count (median of the cell slopes; status inconclusive). At the HDL level the HDL-aware packer's wirelength ratio to the known clustering is 1.099, against the stock packer's 1.219; the direct comparison norns against stock gives wirelength 0.902 (supported), CLB count 0.961 (supported) and critical-path delay 0.954 (supported). The H20 wirelength result is not robust to size: for $B \geq 32$ (n=39) it is inconclusive (geometric mean 1.091, Holm p=0.154). The gain is a leak, not a result about HDL knowledge. By style, norns against stock gives wirelength 0.834 (supported) for dp, where word recovery returns one word per block, so the word affinity file is the block partition on the flip-flops (purity 0.9998 against the stock packer's 0.9501), and 1.000 (not supported) for ctl, where there are no words and 27 of 27 Norns .net files equal the stock files. This round is therefore no evidence of an HDL-knowledge gain beyond the leak. H23 is not supported under the literal rule (median purity-gain difference 0.000 per pair, p<0.001, 51.9% of pairs positive, n=27), but its ceiling makes that outcome close to forced: in 13 of 27 pairs d is exactly 0, because the stock purity of the dp design is already 1.0 (chains at small B and all random graphs), and the mean headroom (one minus the stock purity of the dp design, the largest gain possible) is 0.047, which is about the 0.050 threshold the median must reach. Exploratory: the stock gap sits in the mesh (WL 1.423, CPD 1.184) and the tree (WL 1.522, CPD 1.163); chains (WL 1.004, CPD 0.993) and random sparse graphs (WL 1.019, CPD 1.042) leave almost none. Conjecture, not tested: a chain packs well by connectivity alone, and a random graph has no locality that a clustering could use.

a) What this round establishes.: On clustering-optimum-by-construction designs the stock VPR packer leaves about 22.0% of wirelength and 9.3% of critical-path delay on average at the netlist level (geometric mean over all topologies), concentrated in the mesh and tree constructions (WL 42.3% and 52.2%; CPD 18.4% and 16.3%) and absent in chains and random sparse graphs; the netlist-level wirelength ratio is 1.073 at $B = 4$, peaks at 1.301 at $B = 64$ and is 1.182 at $B = 1024$, so it does not grow monotonically with size. These numbers are against what VPR achieves from the forced block clustering, not against a proven optimum. What the round does not show is that HDL knowledge closes any of it. A non-leaky test of that needs designs where the HDL words do not coincide with the optimal blocks: words that span several blocks, or blocks that mix several words. That is the next question. Limits: the optimum is a clustering optimum by construction, the cut is an upper bound on the minimum cut within the stated slack (and the exactness check is not supported under its literal rule), and ratios are against what VPR achieves from the known clustering, not against a proven placement optimum. The generated functions are random or XOR-heavy, which is not the structure of human RTL (compare the distribution-shift reading of Loki's round three). Sizes stop at 1024 blocks and no dedicated critical-chain design was built.

VI. Conclusion

Seeding VPR's anneal is worth it only as a calibrated cold start: at VPR's own start temperature the cluster layouts and the provenance hierarchy place at parity with the analytic start, but from a good final placement they keep winning when the anneal is restarted hotter, up to the largest tested scale that still wins – a lower bound on the fall point where the sweep hit its cap. The recipe is that crossover table, not a seed-specific tuning: start

at or below it, and expect the seed to inform the coarse placement while the anneal supplies the fine one.

Round two adds bitwise seeds to the provenance hierarchy. At VPR's own start temperature the best of the four new families, words_partial, is 0.994 of the baseline's wirelength; that is a figure selected as the best of four on the data it is reported on, it sits inside the 0.980 to 1.029 band that every soft arm falls in, and it is not an effect. The question that made the idea worth testing, bit-level structure against module structure, comes out inconclusive (H7a) and not supported (H7c) on these designs.

Appendix

This log records the thought pattern of the study so a reader can follow how each result produced the next question. Entries are dated; numbers are those of the generated tables.

a) 2026-10-23, round one: the question and its hypotheses.: Does seeding VPR's anneal from netlist clusters or the provenance hierarchy improve placement, and how hot can the anneal start before it destroys the seed? Hypotheses: a seed helps at VPR's own start temperature; the helpful temperature range is bounded by a design-level crossover; the provenance hierarchy and recovered clusters differ; hard regions help or hurt.

b) 2026-10-24, what round one said.: Soft seeds sat at parity with the analytic baseline (wirelength 1.009 full, 1.015 inter-cluster only; channel width 0.973; runtime 1.087); hard regions were worse (1.088, 14 of 104 better); seed retention was 0.68/0.70 at scale 1 but a seeded run won only 57 of 105 head-to-heads at twice the crossover; Urd was the worst soft and the best hard seed.

c) 2026-10-24, what it suggested.: The baseline is VPR's connectivity-aware analytic start, so better than random was never tested; the anneal budget may erase what a seed adds (an effort curve is untested, N3); VPR's flat placement may already rediscover the structure the seed carries (N1); the designs are small, median 125 CLBs (size effect untested, N5); and whether the gain is the clustering or the coarse layout was not separable (N4).

d) 2026-10-24, Peter's bitwise idea.: "consider that bitwise is more relevant in the Norn path so we should consider that as an additional seeding aspect (not necessarily the entire HDL structure (though hierarchically yes)). Let's include this and run the next set of experiments noting this thought pattern in the appendix of each paper so we can see the progress ... write up into the papers findings." Reading: Norns is the datapath path, bits are the unit of structure there, so word and bit-slice structure is a seeding aspect of its own, with the module hierarchy kept at the coarse level.

e) 2026-10-24, round two: questions and hypotheses.: N7 (bitwise seeds): H7a to H7d. N1: H1 (VPR rediscovers the hierarchy). N2: H2 (structure predicts the gain). N4: H4 (random start, permuted clusters, quotient layout, analytic start). The word membership had to come from data: the provenance BLIF line is a statement, so the membership is read from Odin III's bus names and source spans and joined by net id (Section III-G); only about 49% of the LUTs get a word.

f) 2026-10-24, round two: outcomes.: Inconclusive for H7a, inconclusive for H7b, not supported for H7c, inconclusive for H7d, not supported for H4, inconclusive for H1 and inconclusive for H2; the numbers are in Table XXVI and Section IV-A.

g) 2026-10-24, rules changed after the data.: Two decision rules were touched after the results were visible, and one count was corrected. (1) H2: the rule fixed in the methodology text was "not supported if no correlation reaches $p < 0.05$". When the full data showed one significant correlation, a negative one (best new-family gain against $z(Q)$, $\rho = -0.59$, $p = 0.021$ with the null of the first run), the rule was briefly changed to count only positive correlations, which turned the status into not supported. That change is reverted: under the rule as first written the status is inconclusive. The null was then recomputed without the cap on edge swaps (RQ-N2) and the correlation is $\rho = -0.58$, $p = 0.023$. (2) H7d: the first rule called "at least as good" supported at a geomean ratio of 1 or below, with no test and no margin, and so called H7d supported at 0.980 (better on 9 of 15, $p = 0.607$). A geomean near 1 with a non-significant sign test is no evidence either way, so the status is inconclusive ("no evidence of a difference"). (3) H1: the count of designs above the null in compactness was first divided by the number of designs scored on NMI; it now has its own denominator, and the module row no longer counts the designs with fewer than two real modules (13 designs became 11). The status of H1 did not change.

h) Still open.: N3 (effort curve: how the result changes with the anneal budget, e.g. inner_num); N5 (size: designs of 1000 CLBs and up, where a seed has more to say); N6 (region looseness: constraint regions between the soft seed and the tight regions of round one); LUT coverage of the word membership (a front-end change, not a seeding change); more than one VPR seed per configuration. Next round: H7d is to be tested as non-inferiority, which needs a stated margin on the geomean wirelength ratio; the margin is to be fixed, together with every other decision rule, in a commit before any run, and chosen from the run-to-run spread that several VPR seeds per configuration will measure. No margin is proposed from the present single-seed data.

i) 2026-10-24, round three: Peter's two questions.: "For Norns, one question is does the clustering step of FPGA mapping become the problem - how is the initial seeding for an hdl construct get clustered? We might have to make our version of clustering for this information (how is it done - just with TVpack?). Second, if we map designs and have provenance (including the clustering) what do we learn from the placed (just by aneealing results) can we make some questions like (what is close to one another

TABLE XXV
How much of each design the words_partial seed places. Logic and memory CLBs only (pads and multipliers are anchored in every arm); Seeded = CLBs in a word cluster that VPR's initial placement is given a slot for; Moved = CLBs whose seeded position differs from VPR's analytic start, for words_partial and for the full words seed.

Design	Placed	Logic+mem.	Seeded	Moved (part.)	Moved (words)
ch_intrinsics	243	14	12	11	11
stereovision3	56	15	11	11	12
diffeq2	189	22	16	15	20
diffeq1	300	37	33	33	35
mkPktMerge	503	36	35	25	35
raygentop	691	144	132	132	141
sha	216	142	130	130	142
boundtop	444	55	53	53	53
mkSMAdapter4B	528	130	96	94	124
or1200	1034	254	158	155	247
stereovision0	1073	707	696	696	707
stereovision1	1091	788	786	785	785
mkDelayWorker32B	1176	117	22	18	88
stereovision2	2479	1956	1952	1948	1956
blob_merge	630	494	321	321	492

TABLE XXVI
Research questions and hypotheses of the study with status and the deciding number.

RQ	Hypothesis	Round	Status	Deciding number
R1	soft seed stays at parity	1	supported	wirelength 1.009
R1	hard regions help	1	not supported	1.088 (14/104)
R1	seeds win at 2× crossover	1	not supported	57/105
N7	H7a words beat Urd	2	inconclusive	0.986 (10/15)
N7	H7b slices beat words	2	inconclusive	WL 0.982, ch. 1.033
N7	H7c hier is best	2	not supported	rank 8/11
N7	H7d partial better than full	2	inconclusive	0.980
N4	H4 random < permuted < quotient	2	not supported	0.997/1.020/1.029
N1	H1 baseline already coherent	2	inconclusive	8/11 mod., 9/15 word
N2	H2 structure predicts gain	2	inconclusive	0+/1- of 9 at $p < 0.05$

and using the provenance info (why are things close to one another?)" and "Also, the outputs (words) and the inputs as words also provide hints of where the words are in norns. Let's build our own Norns based packer from scratch (include it in Norns as a copy/modification to Apack to allow us to our controlled experiements)." Reading: the clustering step (VPR's AAPack, not T-VPack) may be where the HDL structure is lost before placement starts; the I/O buses are words too, and say where words sit; and the annealed placement, read with provenance, may answer which constructs end up near each other and why.

j) 2026-10-24, what round two suggested.: Every soft seed stayed within 0.980 to 1.029 of the baseline: the seeds were applied after packing, to whole CLBs, and ABC had already merged cones (only about half of the LUTs get a word), so packing dilution was untested (N8) and word structure at the pad level was unused (N10). The T0 confound of round two (a seeded start also changes VPR's start temperature) is still open for the seeded arms: round three does not remove it. The analytic-start comparison (H9b) addresses it for the packing question, because there the packing is the only difference.

k) 2026-10-24, round three: questions and hypotheses.: N8: H8. N10: H10a, H10b. N9: H9a, H9b, H9c with the Norns packer. N11: H11. All rules, margins and the grid-selection rule are fixed in docs/plan/round3.md, committed as 4fb8b60 before any measurement.

l) 2026-10-24, round three: outcomes.: H8 supported; H10a supported; H9a module inconclusive (pilot-sensitive), word supported, word+io supported; H9b module inconclusive (pilot-sensitive), word inconclusive (pilot-sensitive), word+io inconclusive (pilot-sensitive); H9c module not supported, word not supported, word+io not supported; H10b not supported; H11 inconclusive (word inconclusive, module inconclusive). The numbers are in Table XI.

m) 2026-10-25, round three fix round (after review).: The rule-provenance description was corrected: the entry d805789, called a clarification, is an amendment that changed two H11 rules (label permutations; the 95% exclusion), and two designs had Stage-4 runs finished before it was committed (Method). H11 is now given under the protocol text (no exclusion, 12 of the 15 designs, 1000 pooled permutations) as the primary status: word inconclusive, module inconclusive, overall inconclusive; the status of the first analysis (the sign rule scaled to $\lceil 12n/15 \rceil$ designs, not registered) was word inconclusive, module supported, overall inconclusive, and the amendment's exclusion gives word inconclusive, module inconclusive. The n=11 pass was added for H8, H10a and H11, Holm now keeps $m = 3$ (no status changed), and the positional null uses 200 permutations. The packer got a log-only proposal counter and an integer-parse error; the packing is unchanged (Method), and the per-design matched-entry and proposal tables are in Table XV. Not done: the proposal-counting behaviour (round 4).

n) 2026-10-24, next.: The effort curve (N3), larger designs (N5), a seed-selector change and a larger gain scale in the Norns packer, and the packing-time cost of the affinity term. Added 2026-10-25: fix the proposal counter (count queued molecules only; return a bool from the candidate-queue step) and the scan order of the group members, then rerun the packing experiments.

o) 2026-10-25, round four.: Peter (2026-10-24): "Queue those as round 4 after round 3 finishes - we're trying to make these better - I like the Loki ideas so yes (especially the idea of patterns from our data (what about the bigger designs, do they provide other insights of patterns). Norns I like, but I'm not sure this machine can handle the big designs. Also, I feel as the design gets bigger it's inevitable that the size of the chip is

related to hierarchy in the hdl. So, I would like to see how the new clustering methods impact results. AND getting some knowns from the dataset we have." Peter (2026-10-25): "So for Norns in the clustering packing phase is there a statement of what gets clustered that could be stated from what the packer does (the provinance) and what the starting design was (this is an answer to the heuristic approach that currently exists in the VTR packer (my hypothesis is that due to the pin width there is some sharing especially of certain paths, but there is also two orthoganal paths (or more) that should be packed together)) - note there might be a group of these statements where 10% are of this type, 75% are of this type and 15% are the leftovers that are non-critical - again the criticalness of the choices is a big factor, and the multi-near critical paths impacts the algorithm. What can we say about that? In both cases is there data on a perbenchmark setting using the provenance information? For loki, I'm wondering if we could have sentences such as why I can't recover this since we know we map it, there has to be some step that fools the unmapping - what are those steps and back it up with data." What round three suggested: the Norns packer raised purity but not wirelength, and the proposal counter and the affinity join were suspect. Questions: N17 what gets clustered and why; N12 knowns; N13 hierarchy against chip size; N14 to N16 the packer, a timing-from-provenance estimate and a stage-by-bit seed with critical-path delay everywhere. Disclosures (entry of the fix round in the protocol): small test runs on ch_intrinsics preceded the gates and were deleted (no result uses them: every census and arm record is dated after the gates passed, and the rule scripts were last edited before); four affinity-join passes and one restarted gate ran for builder and compare bugs; the first dated entry amended N16 items 5 and 6 before any N16 data and the final round-four affinity files (adders added) postdate it. Outcomes: Table XIX. Protocol fbddfa0. Next: fix the seed selector and scan order, rerun round three's purity claims with the adder-complete files, and larger designs.

A. Round five: research log

a) 2026-10-26, round five.: Peter: "Tpeko tells us what is still possible, so from a clustering perspective it's an interesting approach." and "So, the rose and Thomas just taught me stuff. We could build it again, but it's just insight." Peter (2026-10-26) then asked for a netlist-level arm (pure TPEKO, exact optimum) beside the HDL level, reported separately. Question N Q1.6 (area, speed, wirelength); protocol 7886f49, with dated entries: (a) generator changes found by the keep check (outputs, chain I/O, rand cycle, ctl functions, viability filter, spec labels for port-named flip-flops), (b) the netlist level, (c) the netlist check rule (local-search slack, changed after seeing three checks), (d) ctl scalar ports (a word was found where none was intended), (e) exploratory analyses after the data, (f) review fixes (the literal pre-data rules for H18 and H28 made primary, H21 Holm over four tests with 10000 permutations, local-search convergence, the leak). Disclosures: HDL generation and synthesis preceded entries (a) to (d); a pilot VPR run of the two smallest designs preceded the batch and is included; the inconclusive case of the H22 rule test in tests/test_r5.py was rewritten after the data (its first input was "not supported" by the written rule; the rule did not change); the netlist check rule of entry (c) and the label rule of entry (a) were changed after data and are reported as variants beside the literal rules. Research question N Q1.6 (area, speed, wirelength): on designs with a known optimal clustering, how far are the VPR packer and the Norns packer from it, and how does the gap grow with size? The answer is in Table XXIII (statuses), Table XXIV (gap per size) and the discussion; the RQ table of the research log gains the row

Regeneration: python scripts/write_round5.py from data/analysis/r5-results.json (scripts/r5stats.py); the steps that need runs/ (VPR logs, flat placements, BLIFs) are r5.py synth, words, arms, attr, hops, and their outputs are committed in data/r5/ (keep.json, runs.jsonl, designs.json).

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TABLE XXVII
Round-three research questions and hypotheses with status and the deciding number (extends Table XXVI).

RQ	Hypothesis	Round	Status	Deciding number
N8	H8 packing dilutes structure	3	supported	median word purity 0.63, split 4.00 (n=15)
N10	H10a io-words add coverage	3	supported	median gain 24.1%, 15/15 designs
N9	H9a module purity up, CLBs within +5%	3	inconclusive (pilot-sensitive)	purity +0.000 (without pilot +0.035), CLB -1.6%, $p < 0.001$
N9	H9a word purity up, CLBs within +5%	3	supported	purity +0.111 (without pilot +0.120), CLB 0.0%, $p < 0.001$
N9	H9a word+io purity up, CLBs within +5%	3	supported	purity +0.104 (without pilot +0.124), CLB 0.0%, $p < 0.001$
N9	H9b module changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.051, 3/15 better, $p = 0.090$; without pilot 1.069, $p = 0.006$
N9	H9b word changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.032, 4/15 better, $p = 0.188$; without pilot 1.053, $p = 0.016$
N9	H9b word+io changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.030, 4/15 better, $p = 0.188$; without pilot 1.050, $p = 0.016$
N9	H9c module+seed beats stock+seed	3	not supported	geomean 1.064, 2/15 better, $p = 1.000$
N9	H9c word+seed beats stock+seed	3	not supported	geomean 1.039, 4/15 better, $p = 1.000$
N9	H9c word+io+seed beats stock+seed	3	not supported	geomean 1.039, 5/15 better, $p = 1.000$
N10	H10b word+io beats word	3	not supported	geomean 0.998, 4/15 better, $p = 1.000$
N11	H11 same-word pairs closer than nets predict	3	inconclusive	median -0.176, $p = 0.017$, n=12; 9 of 12 designs sign-consistent, 12 needed; with the unregistered scaling (10 needed): inconclusive
N11	H11 same-module pairs closer than nets predict	3	inconclusive	median -0.188, $p < 0.001$, n=11; 10 of 11 designs sign-consistent, 12 needed; with the unregistered scaling (9 needed): supported
N11	H11 overall (both relations)	3	inconclusive	supported only if both are; unregistered scaling: inconclusive

TABLE XXVIII
Round-four research questions with status and deciding number (extends Table XXVII).

RQ	hypothesis	status	deciding number
N17	H17a shape (10/75/15)	not supported	medians SC 4.0%, SI+OR 32.0%, SN+FL 64.4%, n=15
N17	H17a literal numbers	not supported	same medians against 10/75/15 ±0.10
N17	H17b packer changes SC share	not supported	median ΔSC -0.003, Holm $p < 0.001$, n=15
N17	H17c critical edges kept together	supported	median ratio 0.652, Wilcoxon $p = 0.027$, n=9 (literal rule; reading with an n<12 clause: inconclusive (n < 12))
N17	H17d series share above the atom-shuffle null (not a clustering-quality result)	supported	median excess 0.072, 15/15 positive; exploratory molecule-unit null: excess 0.023, p<0.001, inconclusive
N13	H13a module count vs grid area	supported	n=15 (M only: depth not measurable from provenance names)
N13	H13b beyond design size	inconclusive	n=15
N13	H13c packer effect vs hierarchy	inconclusive	n=15
N15	H15 depth estimate predicts critical statements	inconclusive	median ρ 0.137, n=12
N14	H14 norns vs stock: wirelength	not supported	geomean 1.033, 4/15 better, Holm $p = 1.000$
N14	H14 norns vs stock: chan_width	not supported	geomean 1.015, 4/15 better, Holm $p = 1.000$
N14	H14 norns vs stock: cpd_ns	not supported	geomean 0.996, 6/15 better, Holm $p = 1.000$
N16	H16 stage×bit seed: norns.chan_width	inconclusive	geomean 0.974, 8/14 better, Holm $p = 0.684$
N16	H16 stage×bit seed: norns.cpd_ns	not supported	geomean 1.013, 6/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: norns.wirelength	not supported	geomean 1.015, 6/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: stock.chan_width	inconclusive	geomean 0.977, 10/14 better, Holm $p = 0.581$
N16	H16 stage×bit seed: stock.cpd_ns	inconclusive	geomean 0.996, 7/14 better, Holm $p = 1.000$
N16	H16 stage×bit seed: stock.wirelength	not supported	geomean 1.005, 6/14 better, Holm $p = 1.000$

TABLE XXIX
Research question added in round five.

RQ	question (measures)
N Q1.6	On designs with a known optimal clustering, how far are the VPR packer and the Norns HDL-aware packer from it, and how does the gap grow with size? (area, speed, wirelength)